

RUL algorithm comparison

Diffusion-RUL

Generated	2026-05-03 08:13:10
Runs included	2
Report figures regenerated	33
Datasets	phm2012, xjtusy

Contents

- 1. Aggregated metrics (all runs)
- 2. Side-by-side comparison: Diffusion-RUL vs Diffusion-RUL

Ablation & configuration comparison

Aggregated across 2 run(s). Means and standard deviations are computed across seeds with the matching run_id prefix. Each metric gets its own chart so direction and scale are unambiguous.

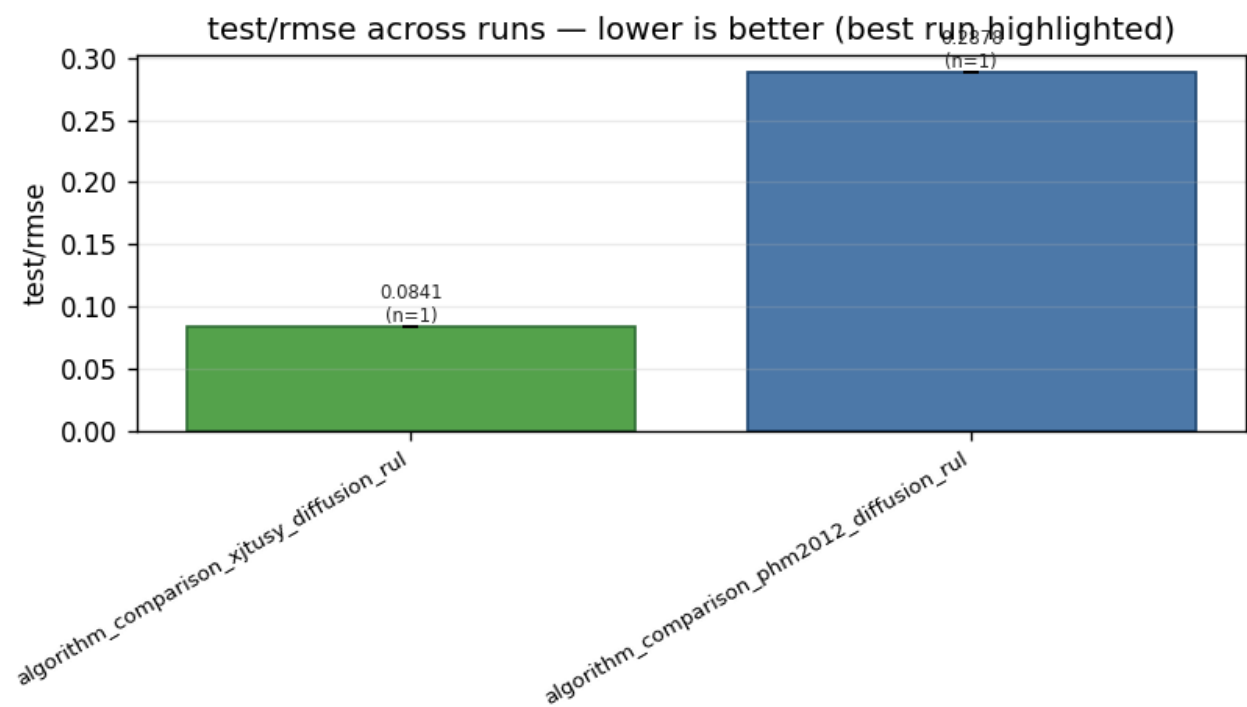


Figure A1. *test/rmse* across runs (error bars: ± 1 std; best run highlighted).

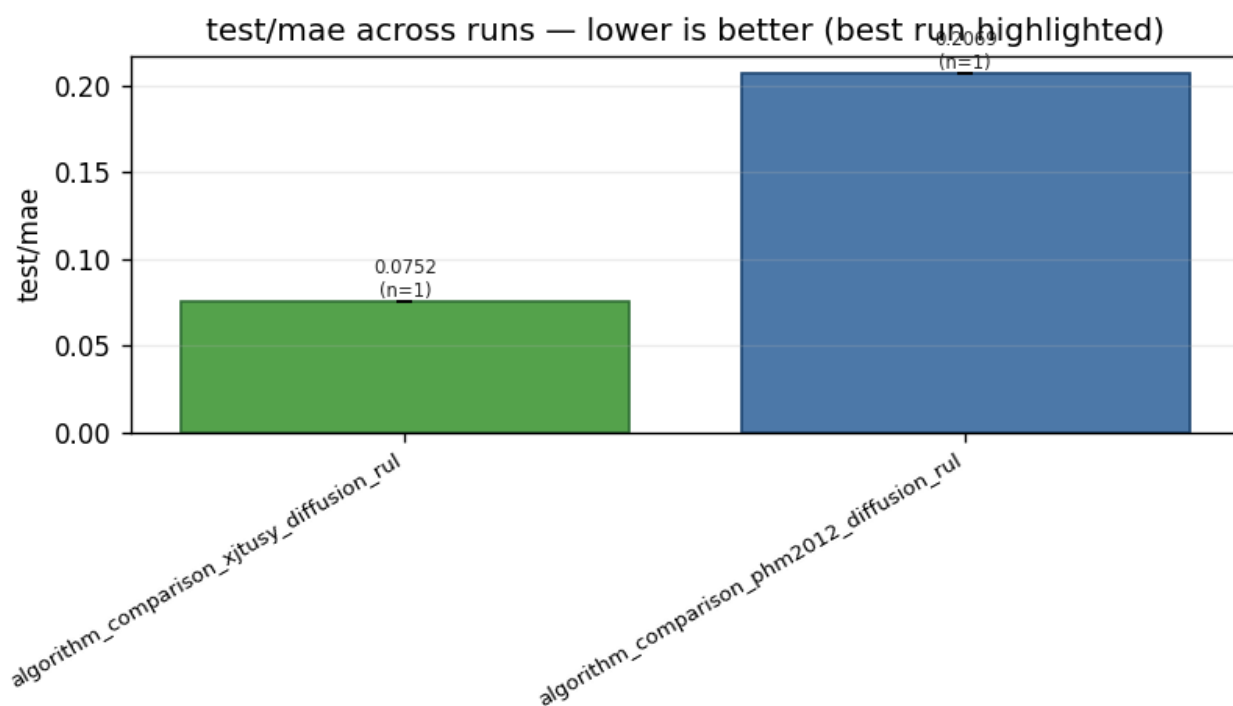


Figure A2. *test/mae* across runs (error bars: ± 1 std; best run highlighted).

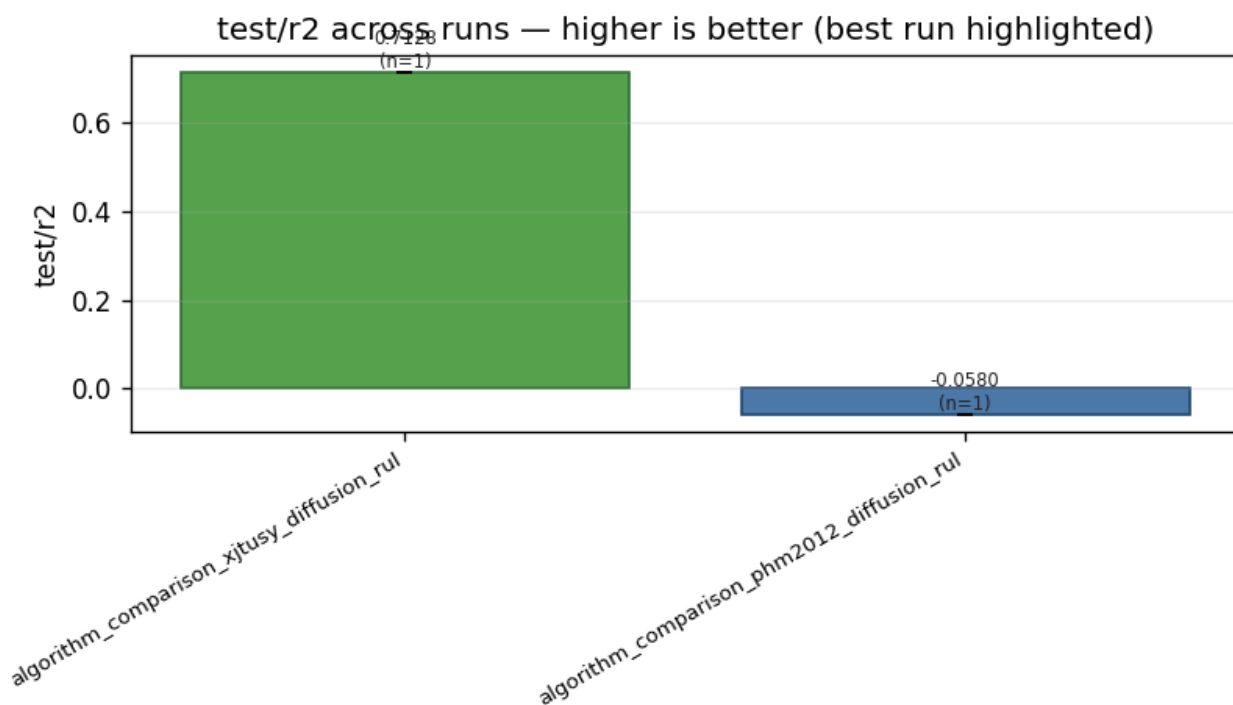


Figure A3. *test/r2* across runs (error bars: ± 1 std; best run highlighted).

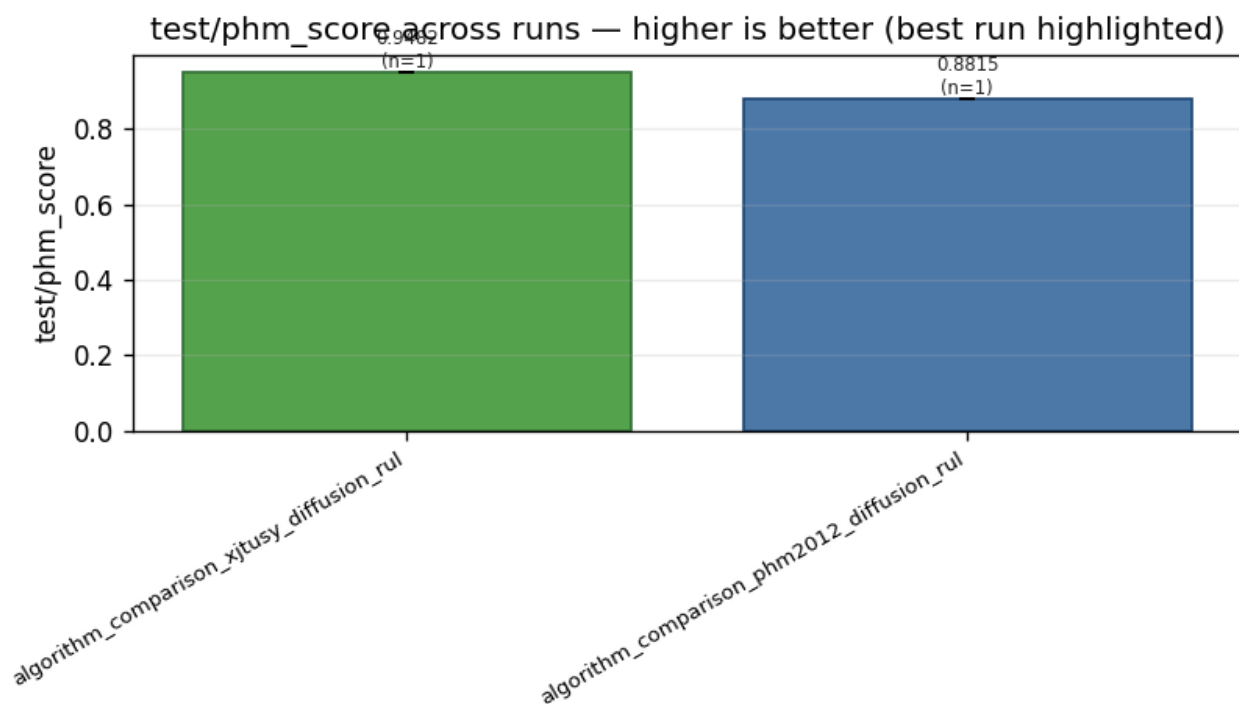


Figure A4. test/phm/score across runs (error bars: ± 1 std; best run highlighted).

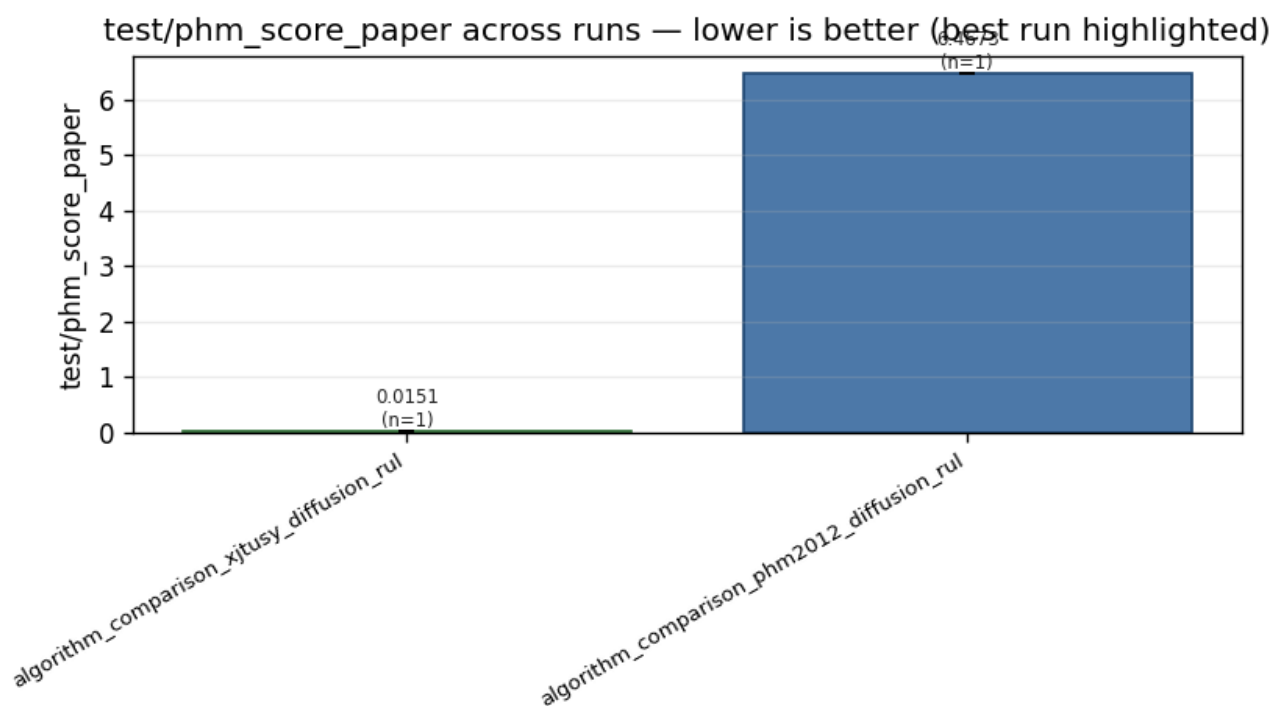


Figure A5. test/phm/score/paper across runs (error bars: ± 1 std; best run highlighted).

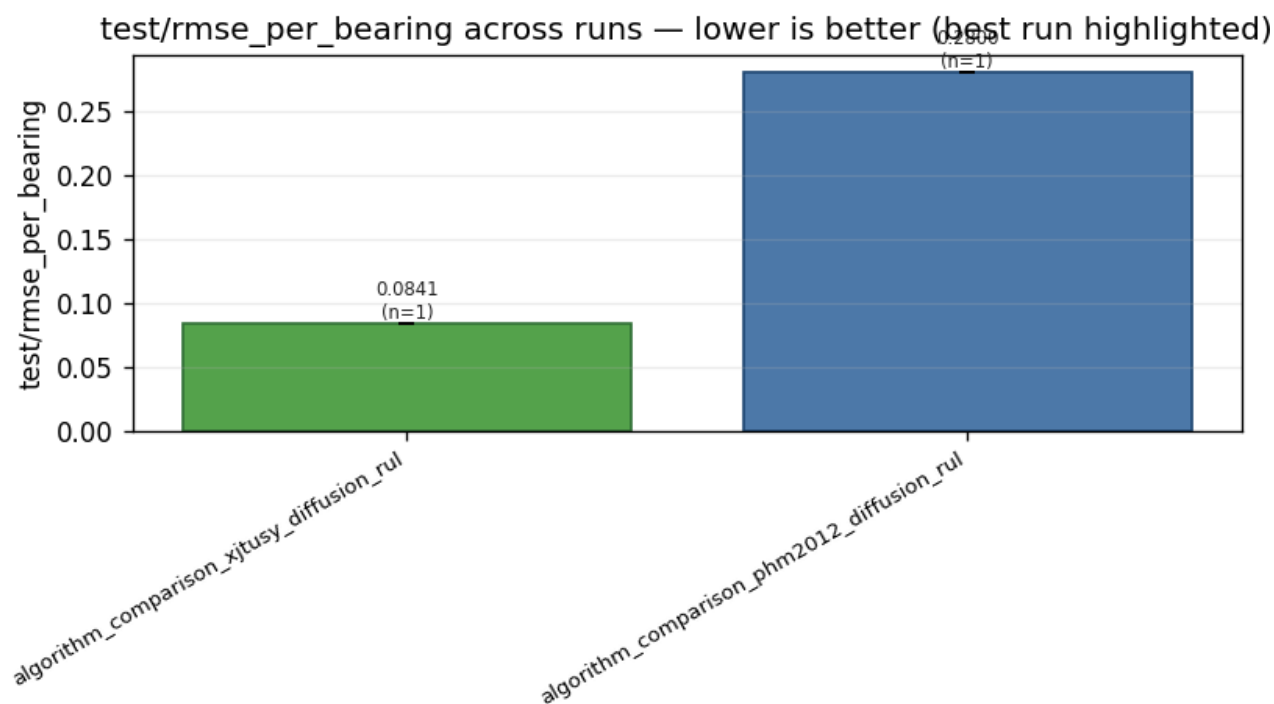


Figure A6. test/rmse/per/bearing across runs (error bars: ± 1 std; best run highlighted).

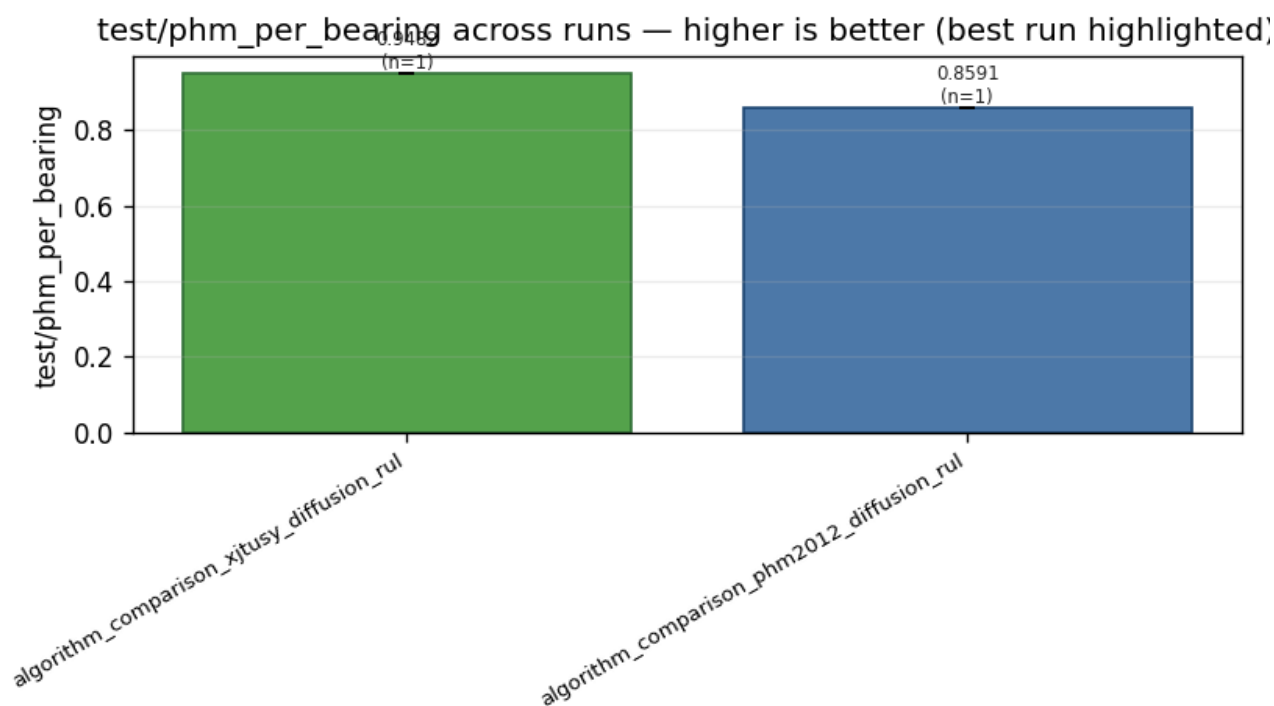


Figure A7. test/phm/per/bearing across runs (error bars: ± 1 std; best run highlighted).

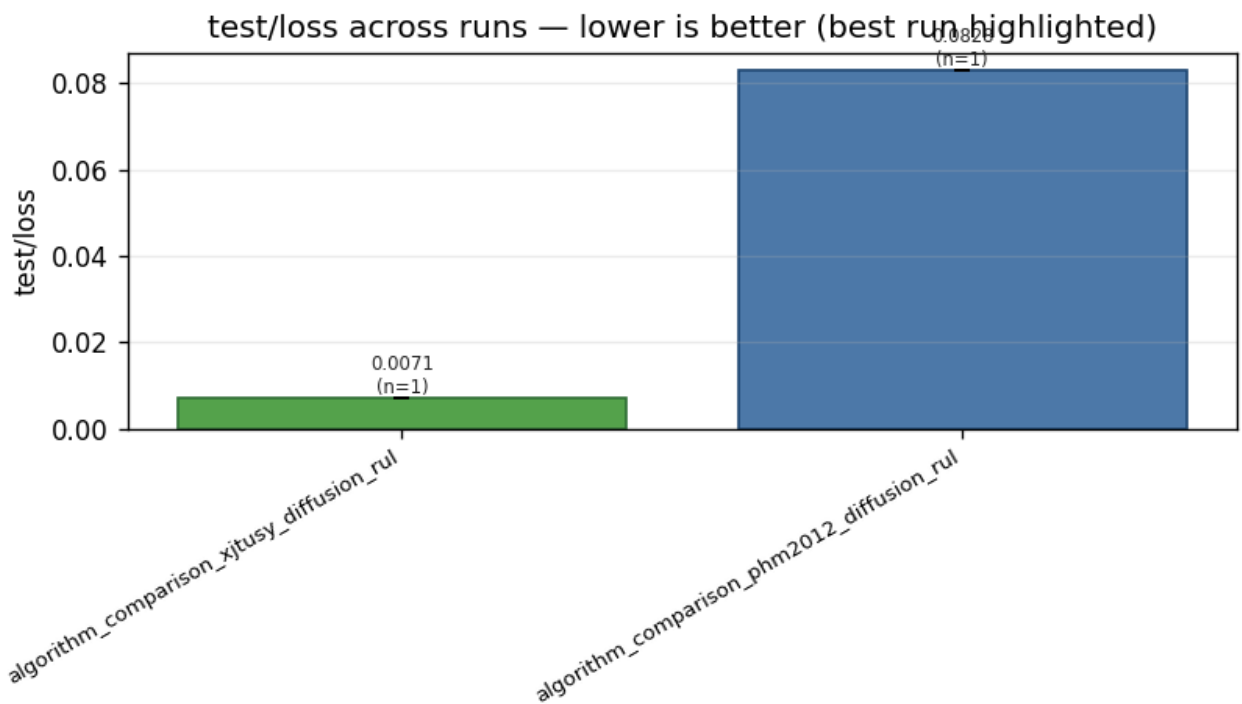


Figure A8. test/loss across runs (error bars: ± 1 std; best run highlighted).

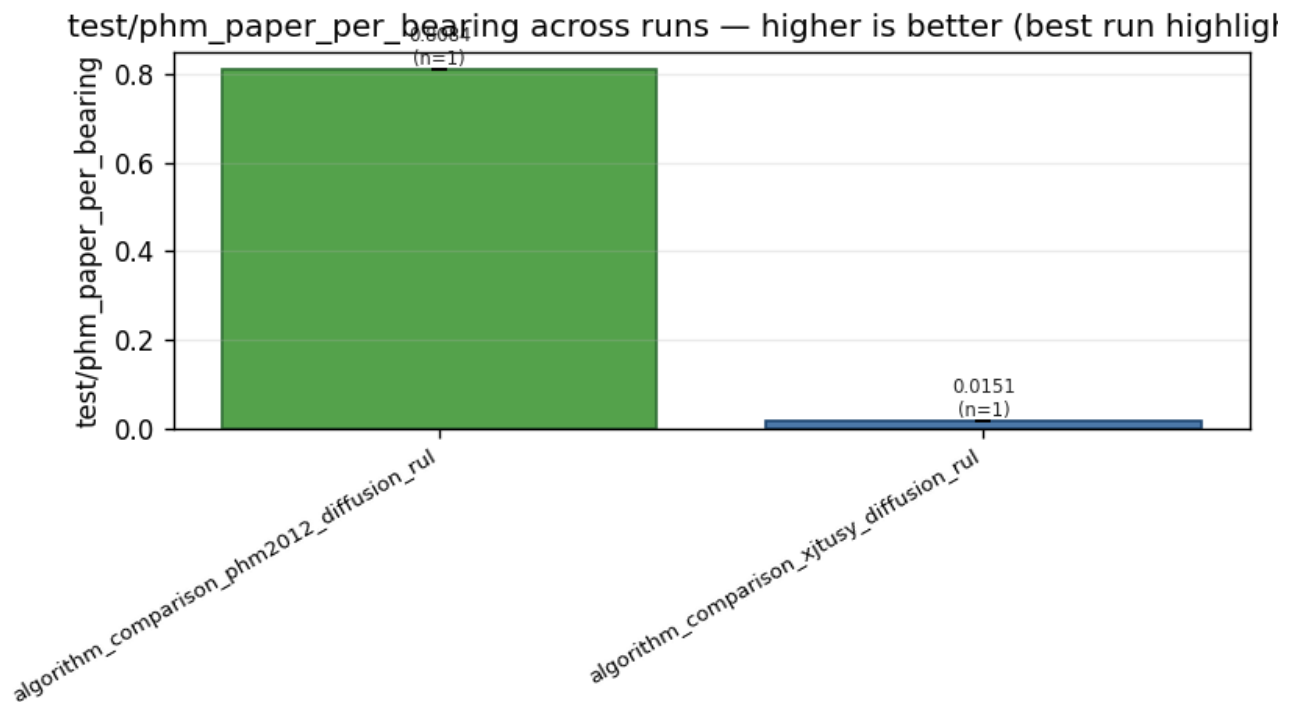


Figure A9. test/phm/paper/per/bearing across runs (error bars: ± 1 std; best run highlighted).

run	test/rmse	test/mae	test/r2	test/phm	test/phm	test/rmse	test/phm	test/loss	test/phm_paper_per_bearing
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[illegible]

Comparison: Diffusion-RUL vs Diffusion-RUL

Same data and training protocol; only the model architecture differs. Recognised model names are ordered consistently for comparison.

Left run	algorithm_comparison_phm2012_diffusion_rul_s42
Right run	algorithm_comparison_xjtusy_diffusion_rul_s42
Left path	/Users/ekkirinaldi/Documents/Git/disertation-rul-prediction/Mamba
Right path	/Users/ekkirinaldi/Documents/Git/disertation-rul-prediction/Mamba

	Diffusion-RUL	Diffusion-RUL
Seed	42	42
Parameters	334,405	334,405
Train time	402.1 s	27.8 s
Dataset	phm2012	xjtusy

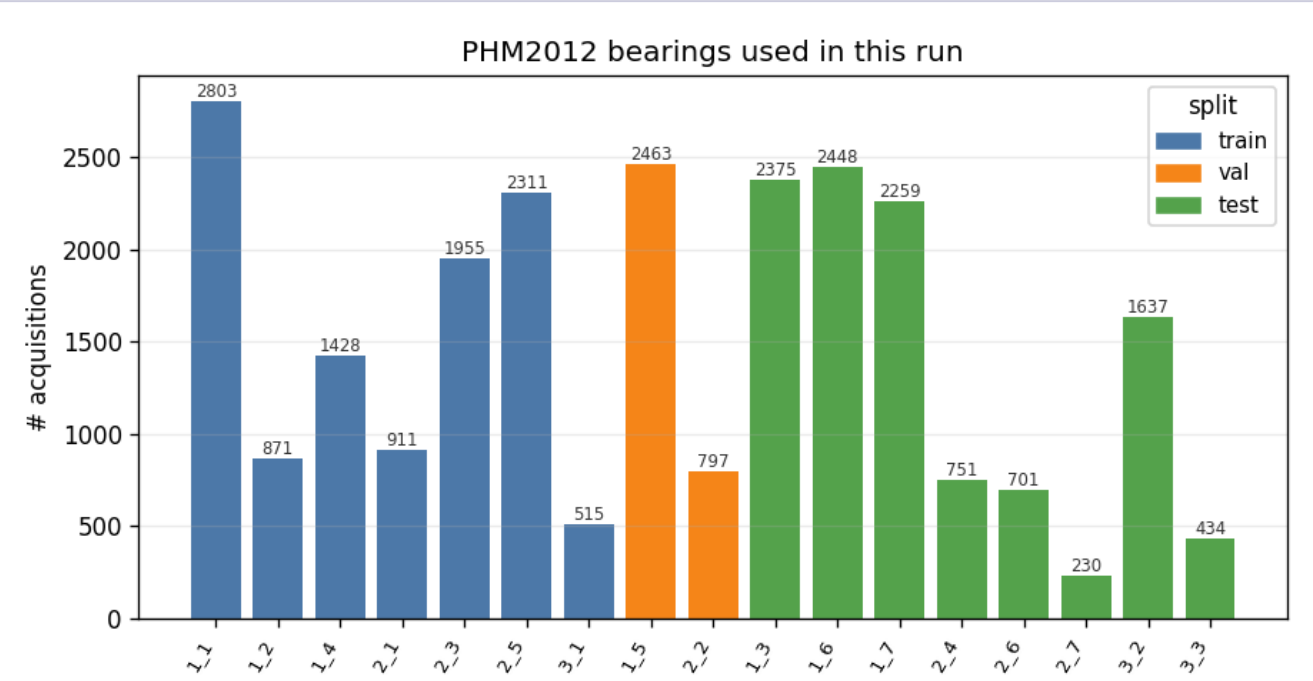
Test metrics

Metric	Diffusion-RUL	Diffusion-RUL	Better
test/loss	0.082846	0.007066	→ right
test/mae	0.206864	0.075225	→ right
test/phm_paper_per_bearing	0.808408	0.015116	→ right
test/phm_per_bearing	0.859137	0.948226	← left
test/phm_score	0.881467	0.948226	→ right
test/phm_score_paper	6.467263	0.015116	→ right
test/r2	-0.058023	0.712818	→ right
test/rmse	0.287830	0.084062	→ right
test/rmse_per_bearing	0.279963	0.084062	→ right

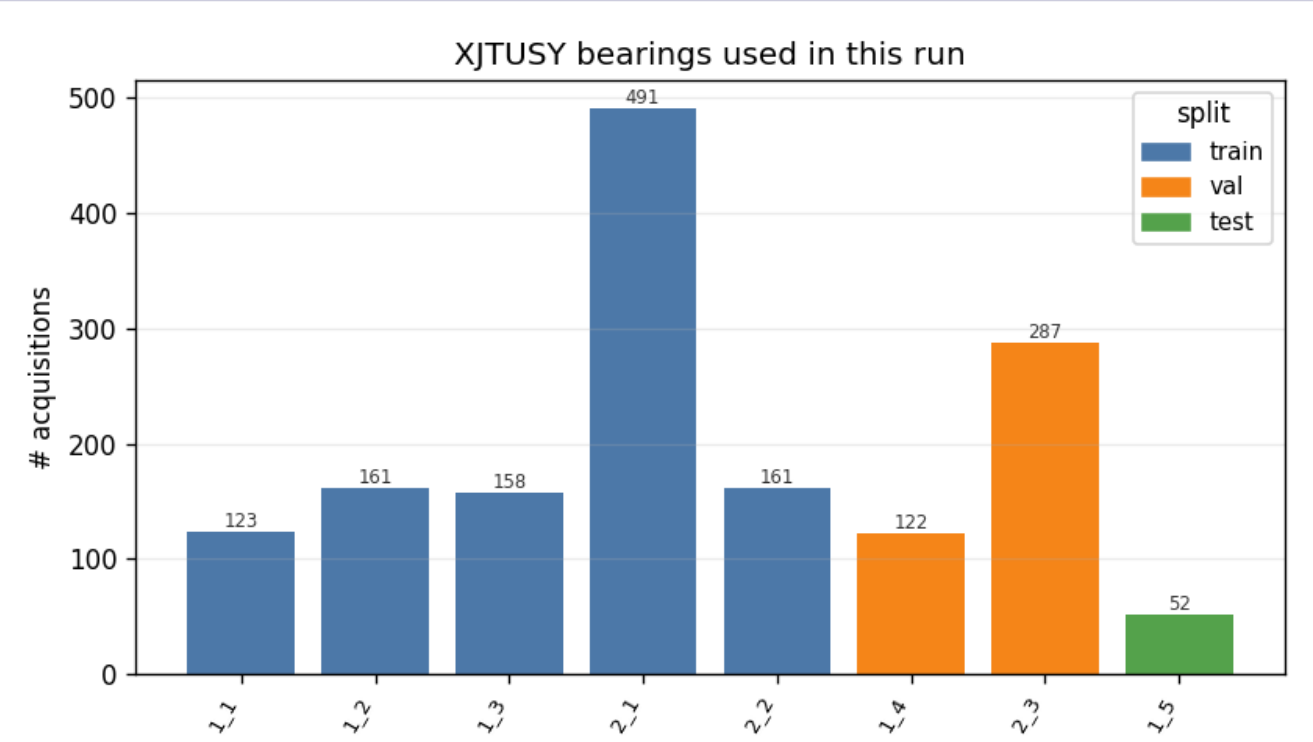
“Better” uses lower-is-better for RMSE/MAE/loss and Liu *phm_score_paper*; higher-is-better for R^2 and project *phm_score*.

Dataset overview

Diffusion-RUL



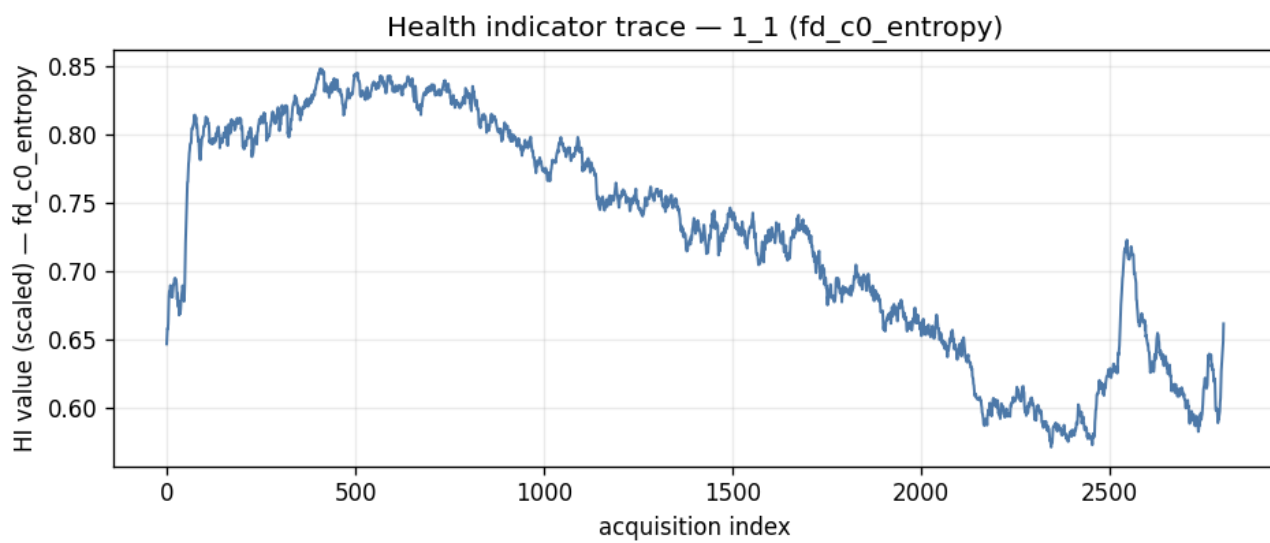
Diffusion-RUL



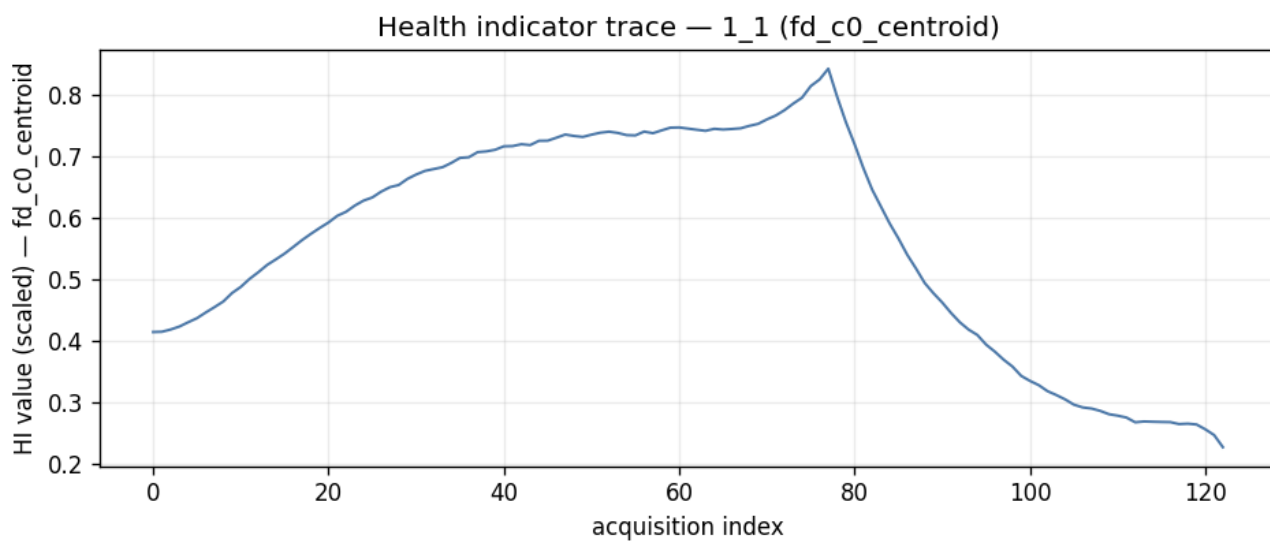
Bearings used (train / val / test) and acquisition counts per bearing.

HI trace (sample training bearing)

Diffusion-RUL



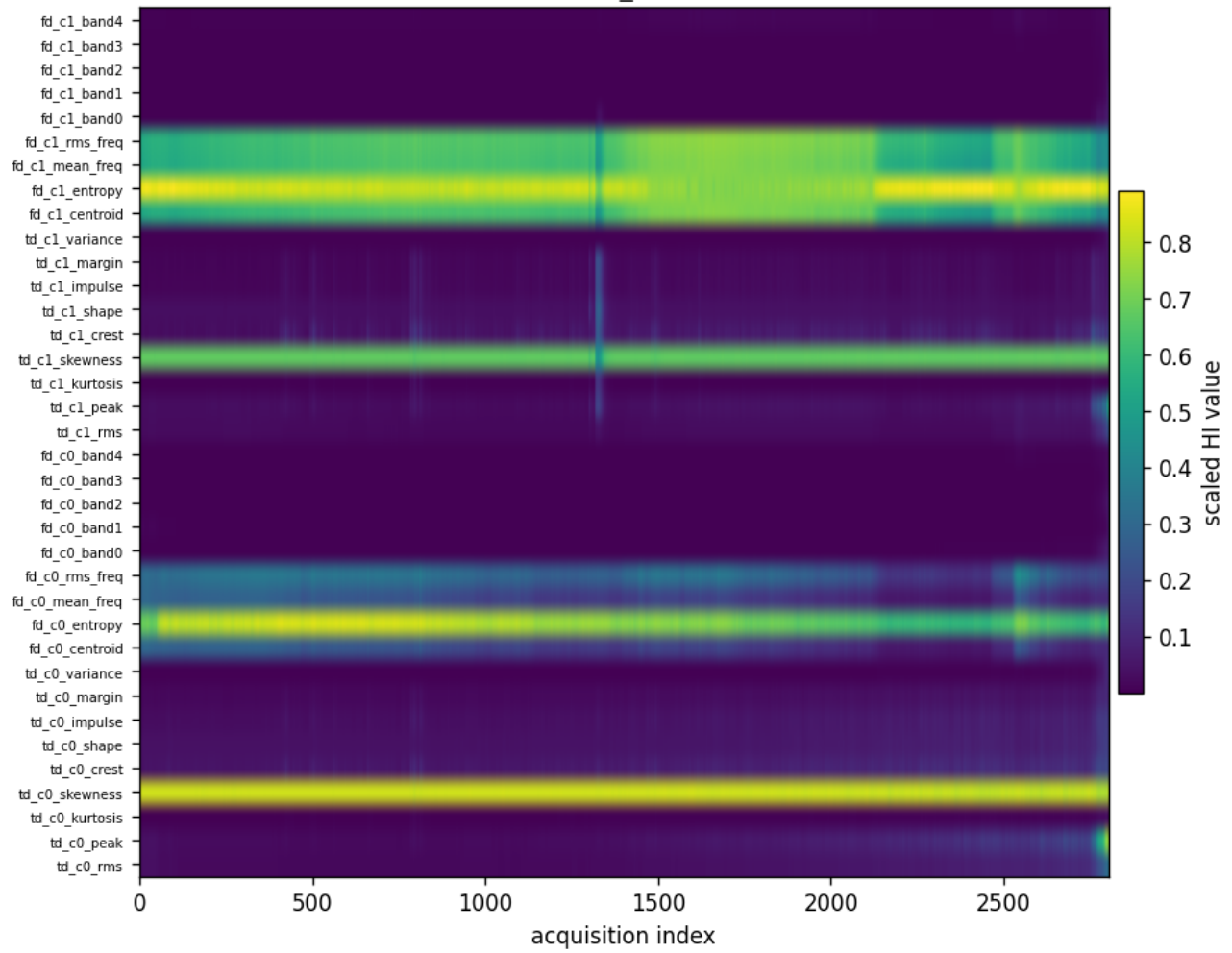
Diffusion-RUL

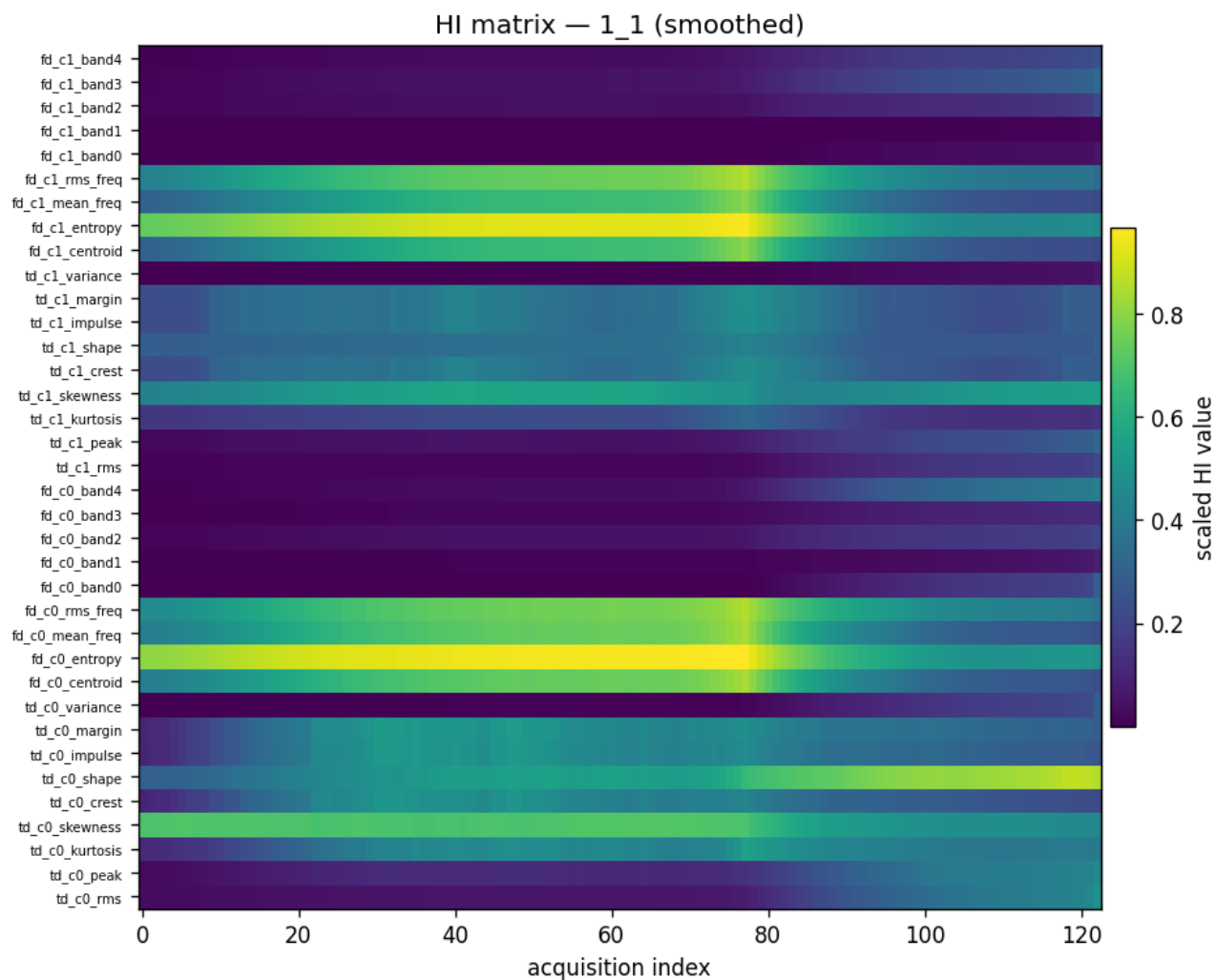


HI heatmap

Diffusion-RUL

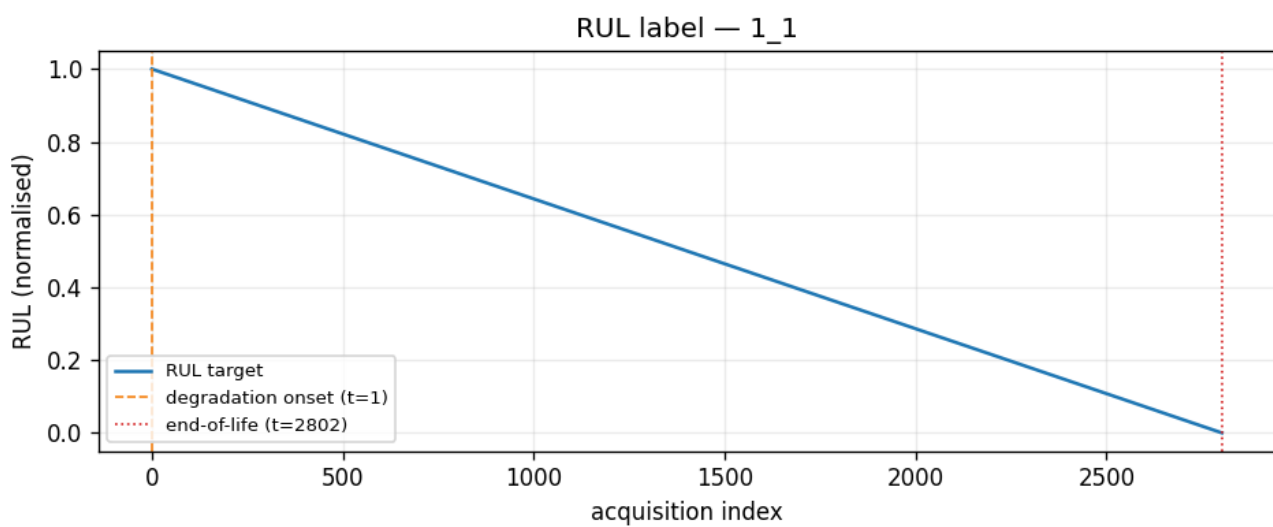
HI matrix — 1_1 (smoothed)



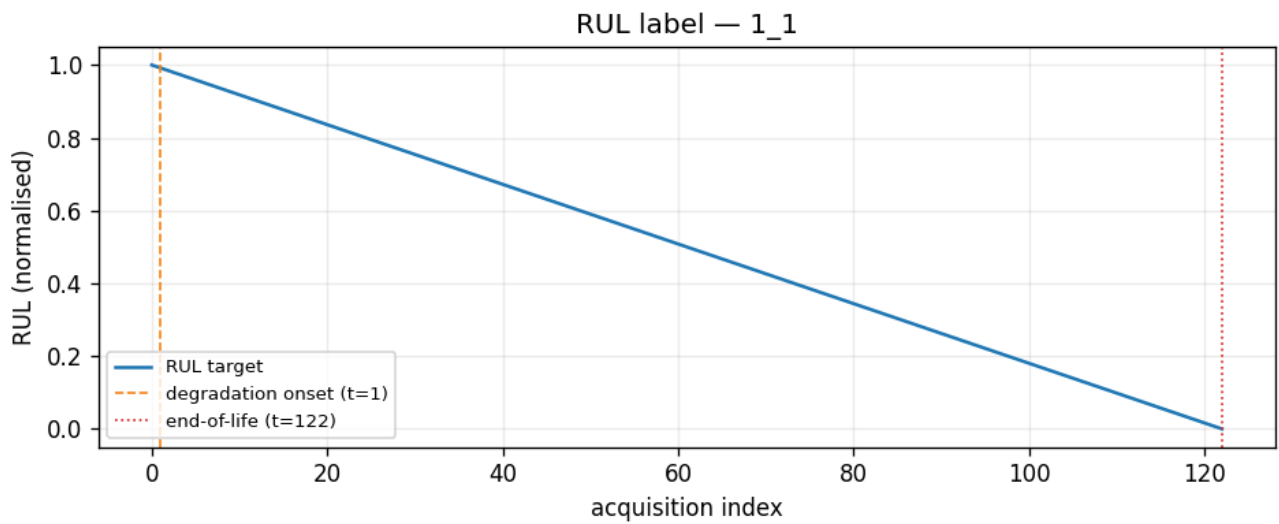


RUL labels (per training bearing)

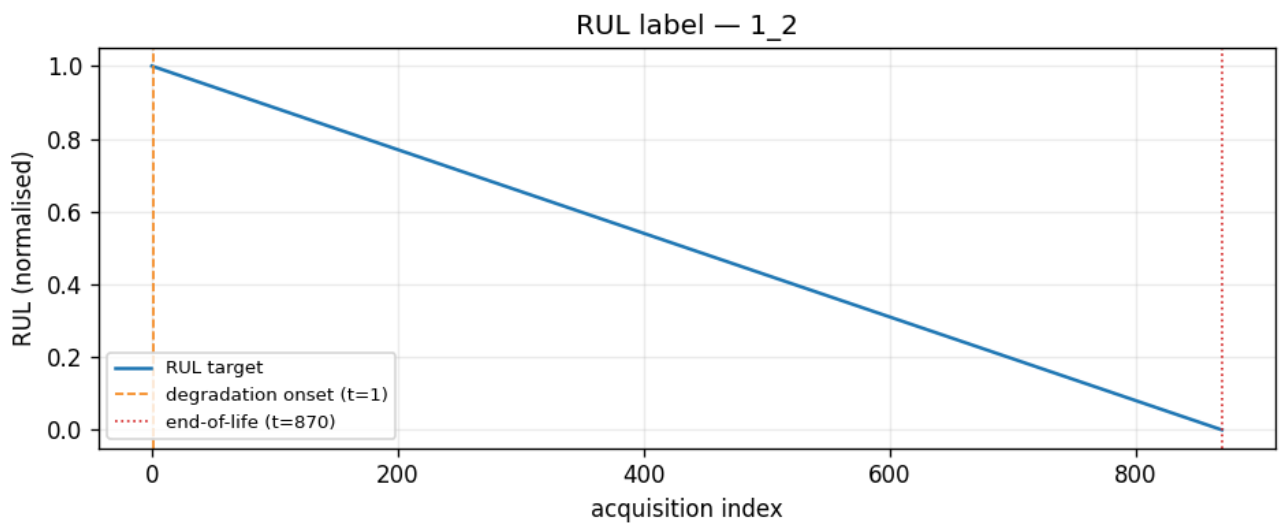
Diffusion-RUL — 1_1



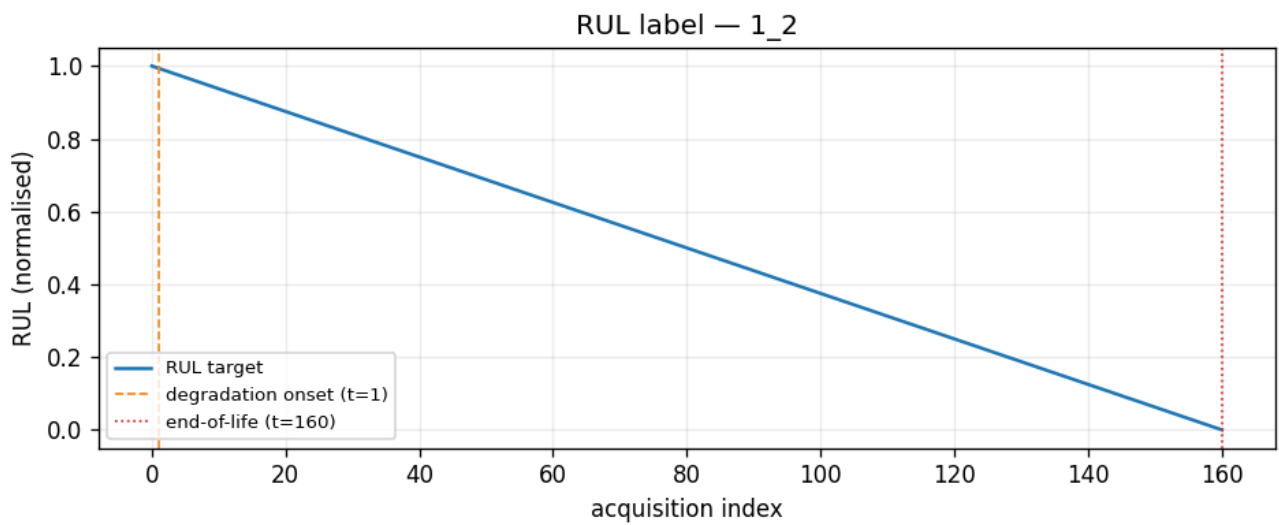
Diffusion-RUL — 1_1



Diffusion-RUL — 1_2

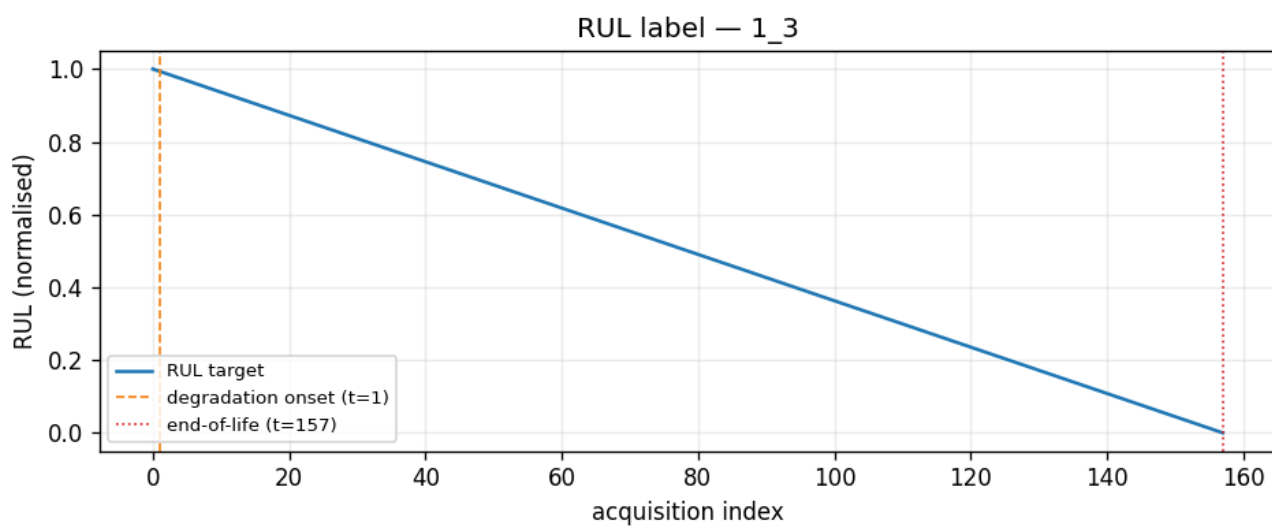


Diffusion-RUL — 1_2

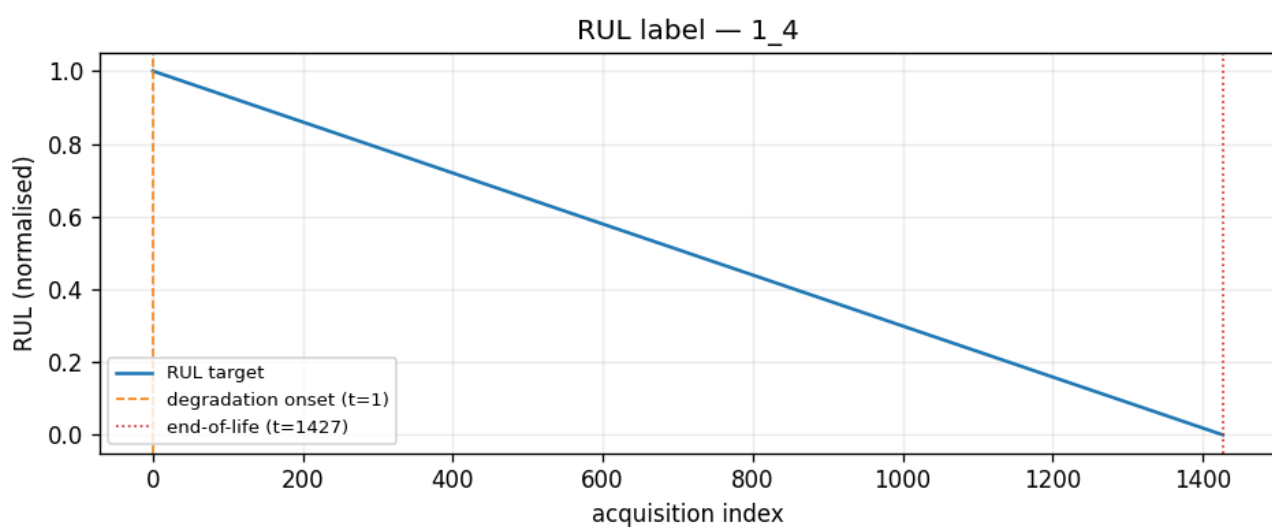


Diffusion-RUL — 1_3

Diffusion-RUL — 1_3

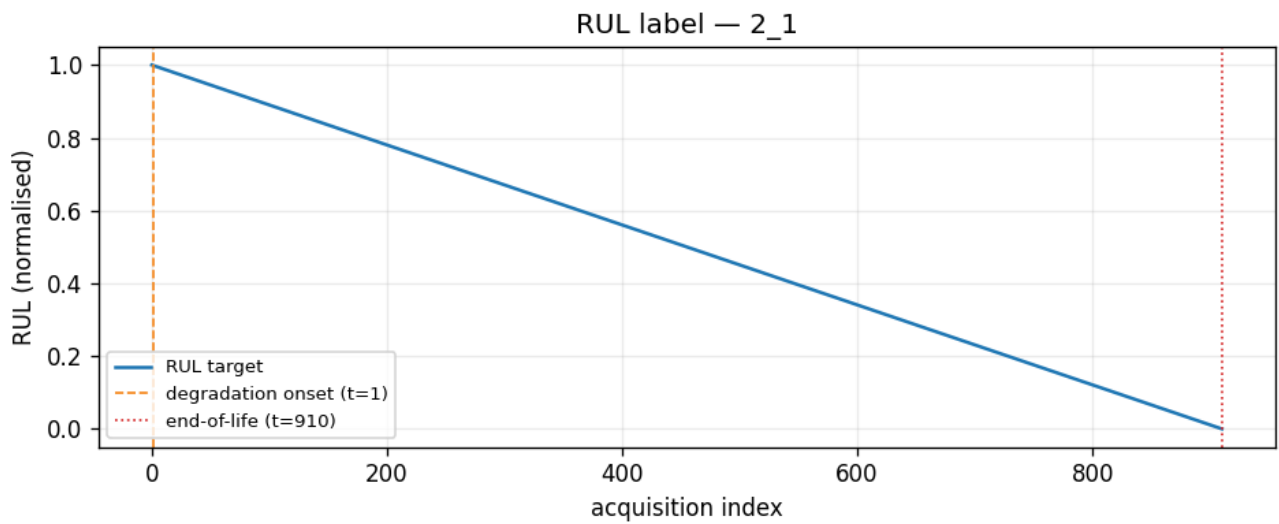


Diffusion-RUL — 1_4

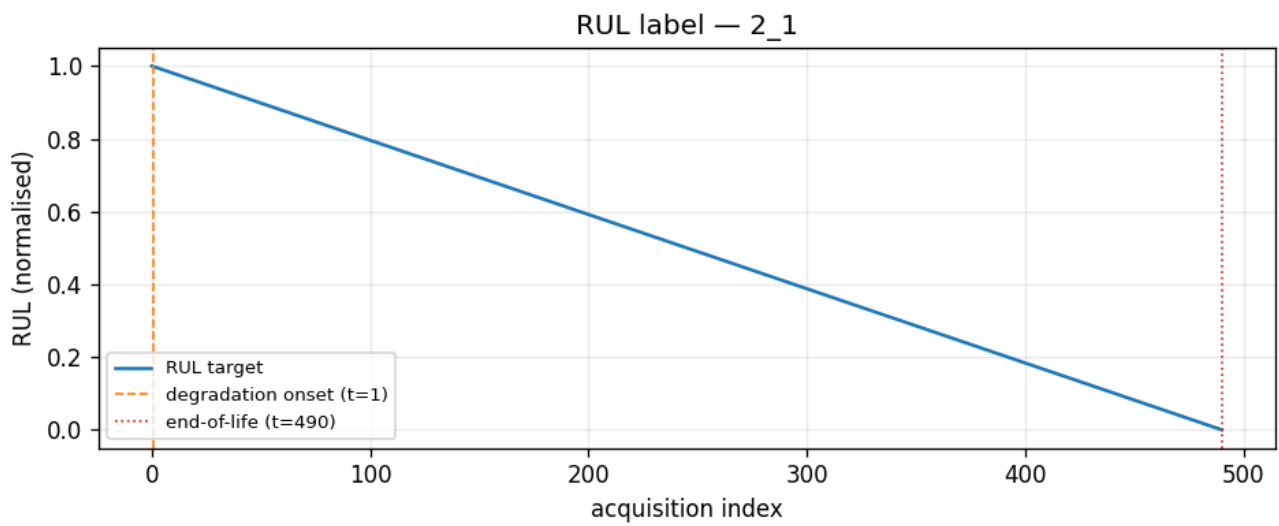


Diffusion-RUL — 1_4

Diffusion-RUL — 2_1



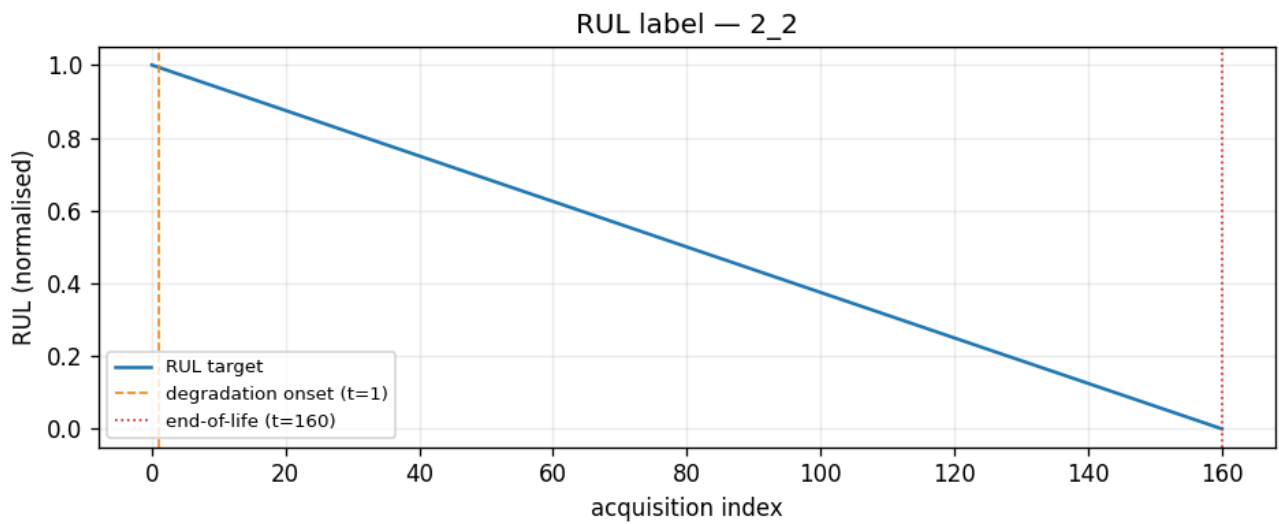
Diffusion-RUL — 2_1



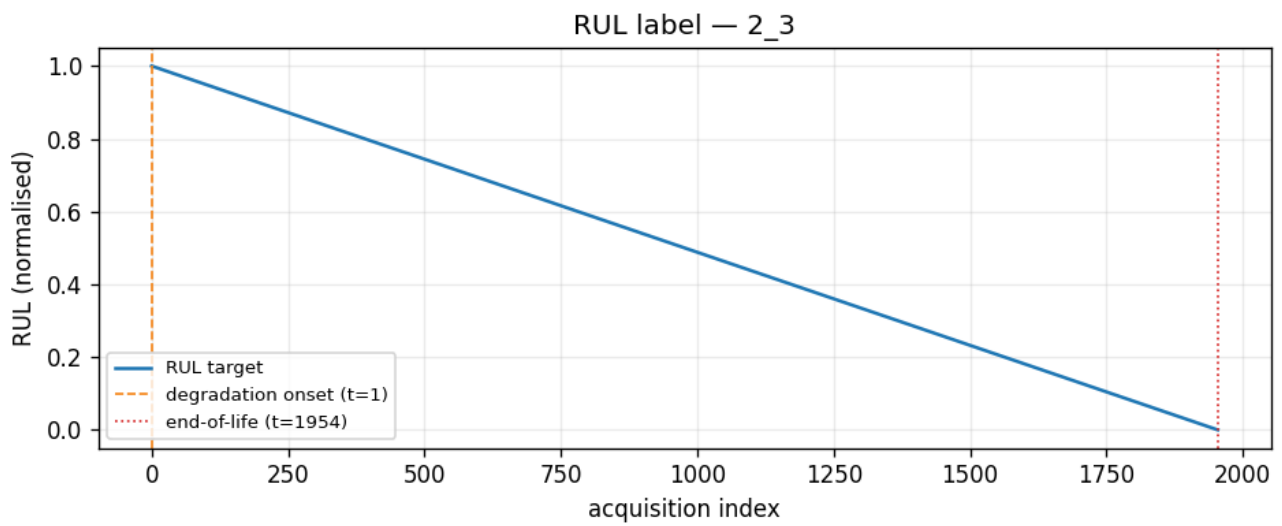
Diffusion-RUL — 2_2

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Diffusion-RUL — 2_2



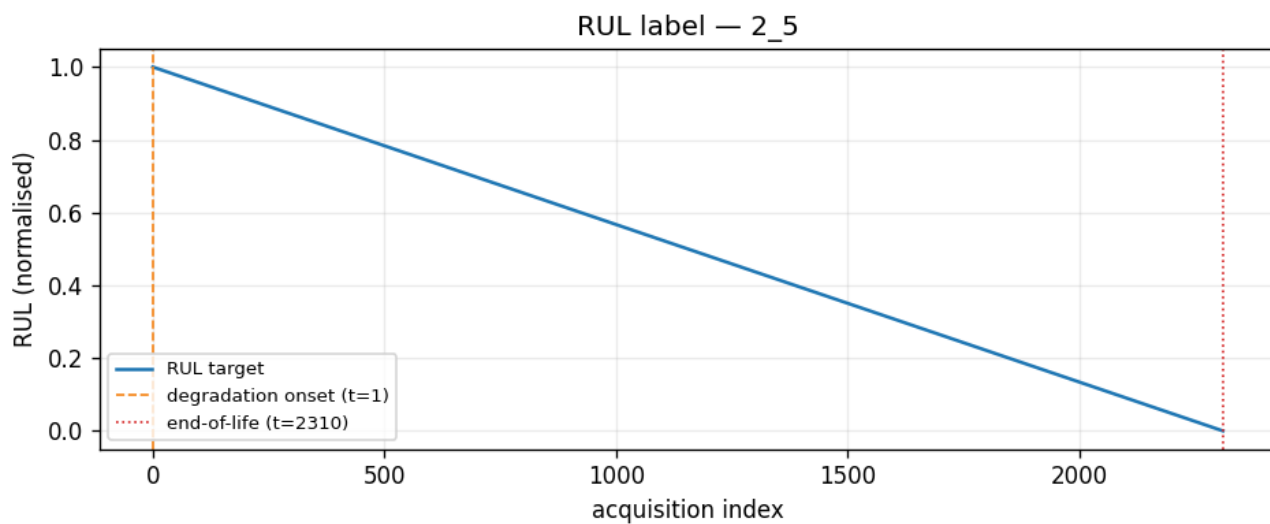
Diffusion-RUL — 2_3



Diffusion-RUL — 2_3

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Diffusion-RUL — 2_5

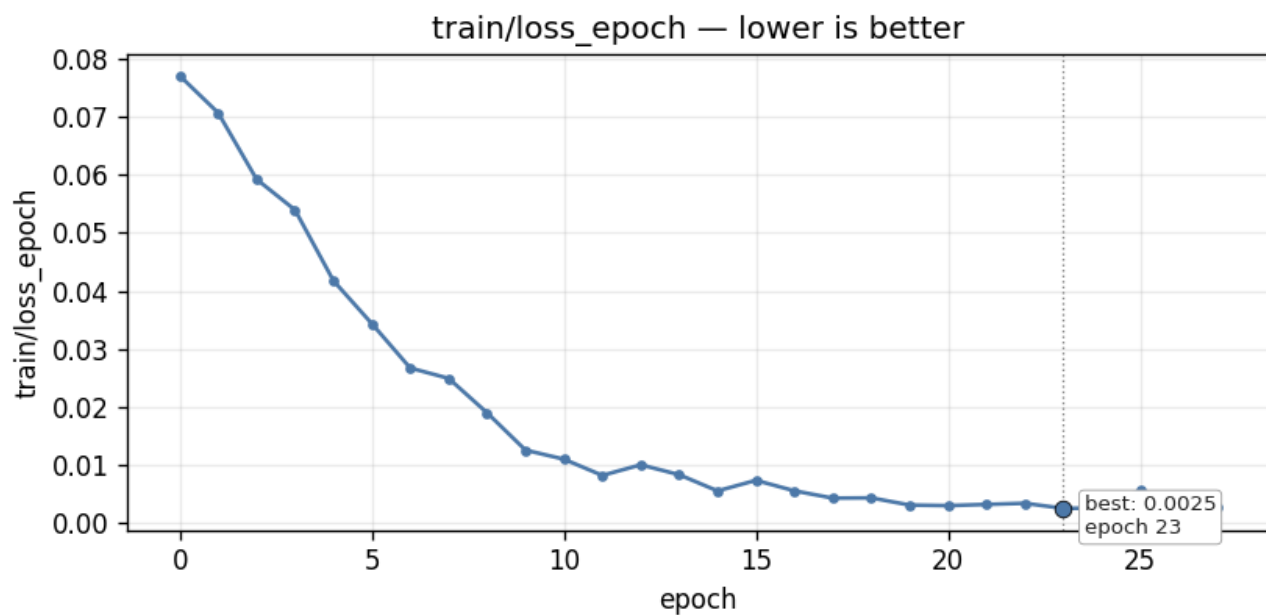


Diffusion-RUL — 2_5

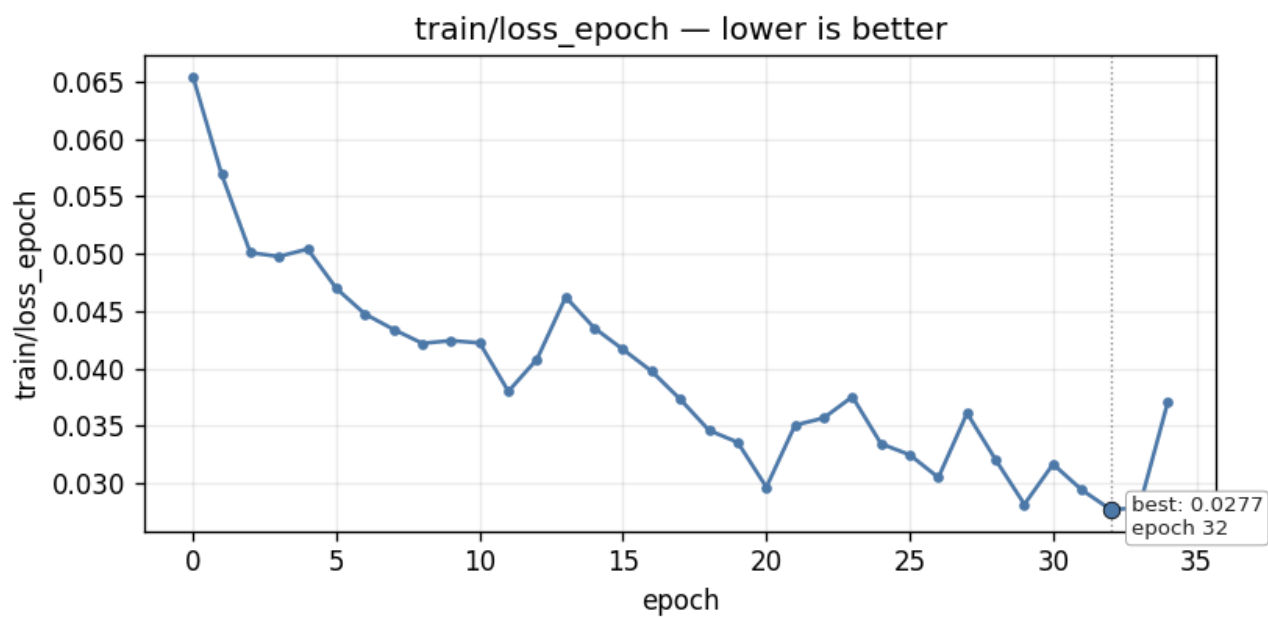
Training dynamics

train/loss/epoch over epochs

Diffusion-RUL

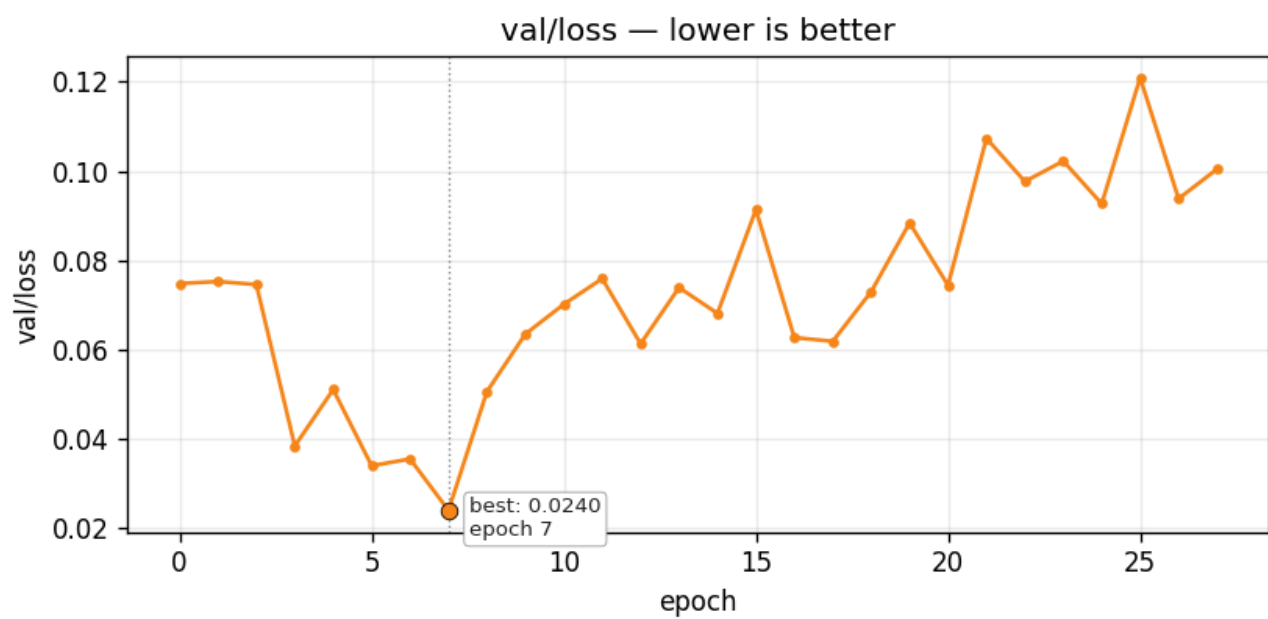


Diffusion-RUL

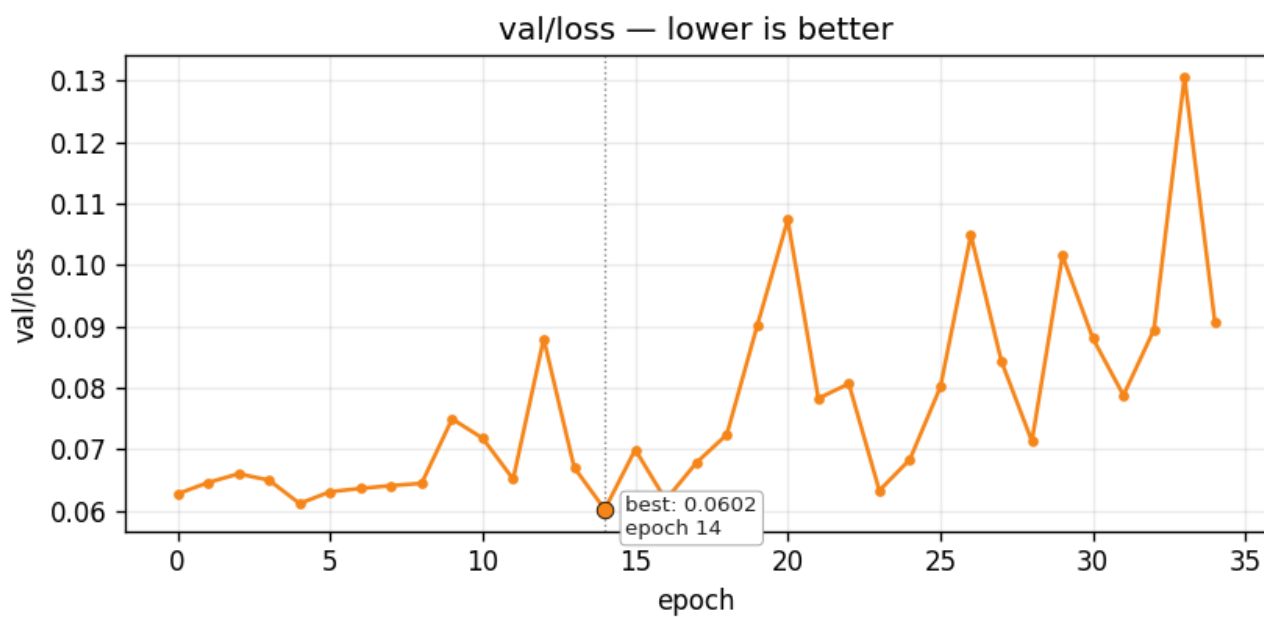


val/loss over epochs

Diffusion-RUL

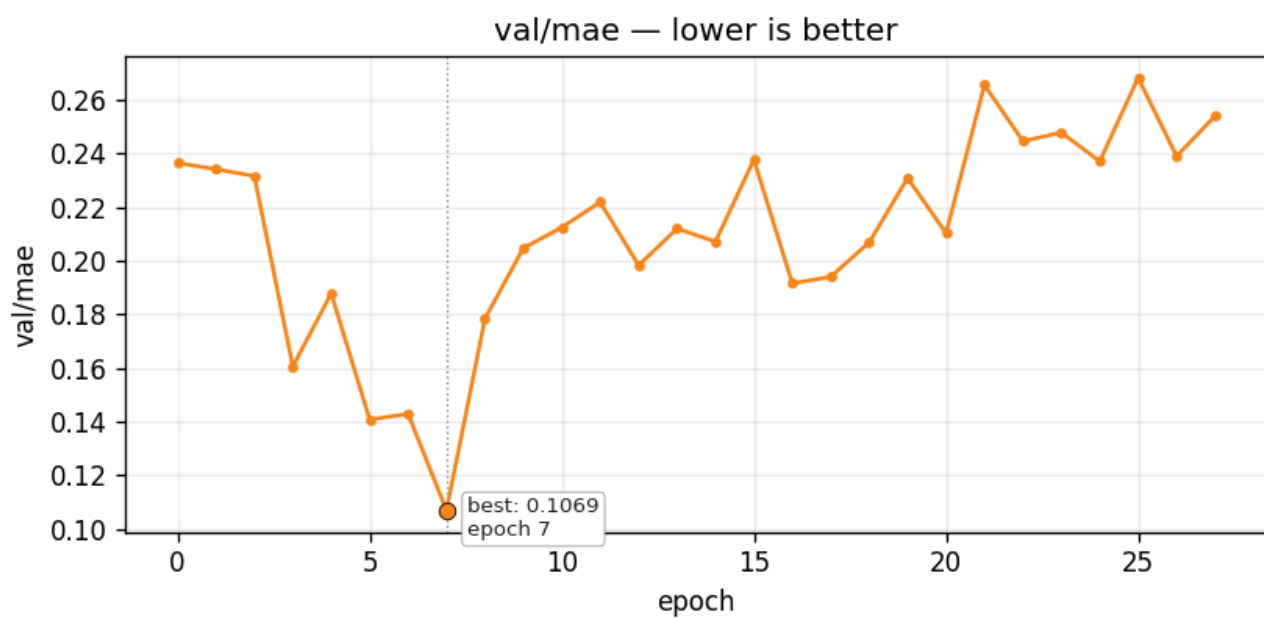


Diffusion-RUL

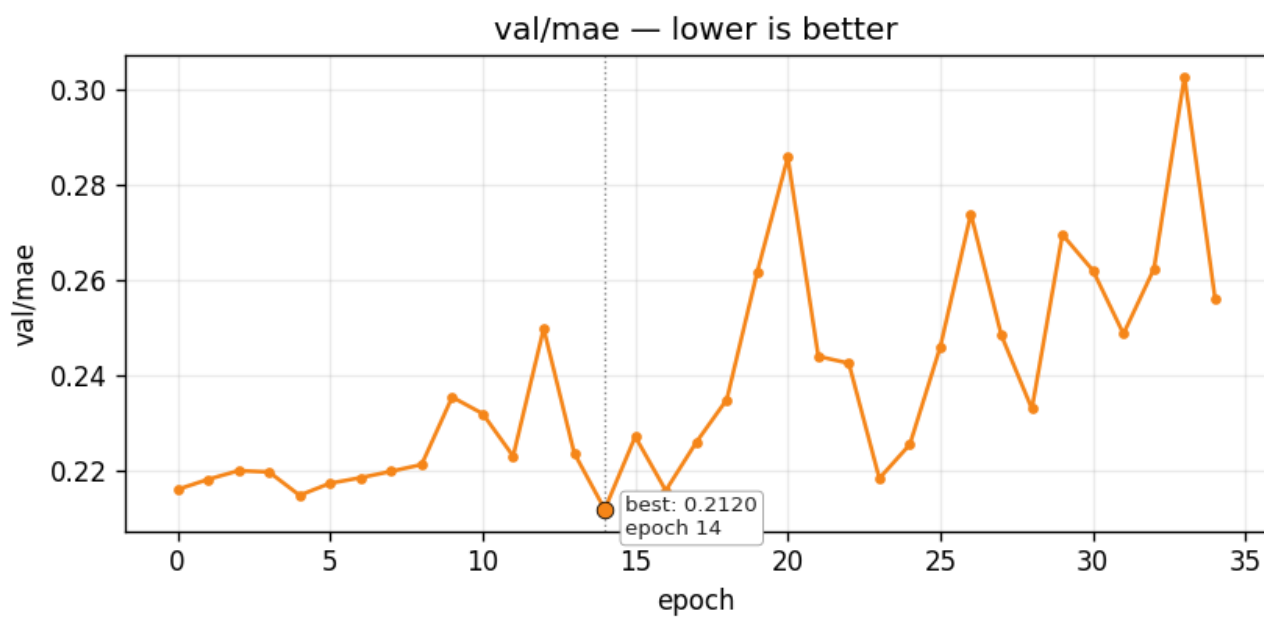


val/mae over epochs

Diffusion-RUL

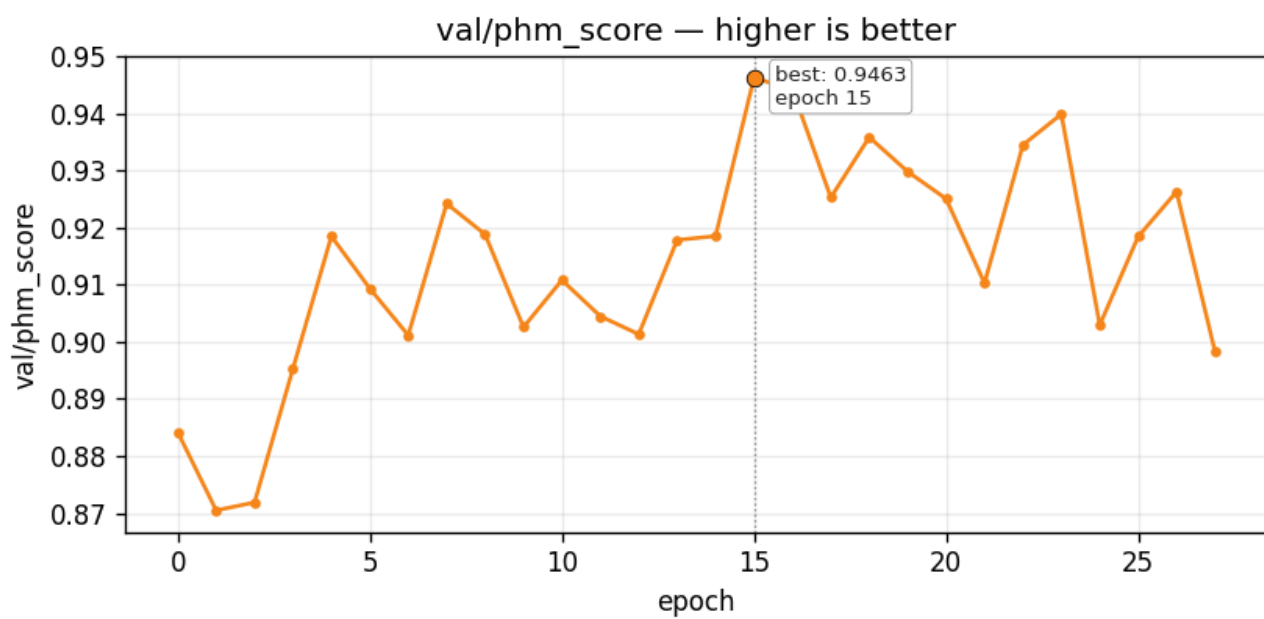


Diffusion-RUL

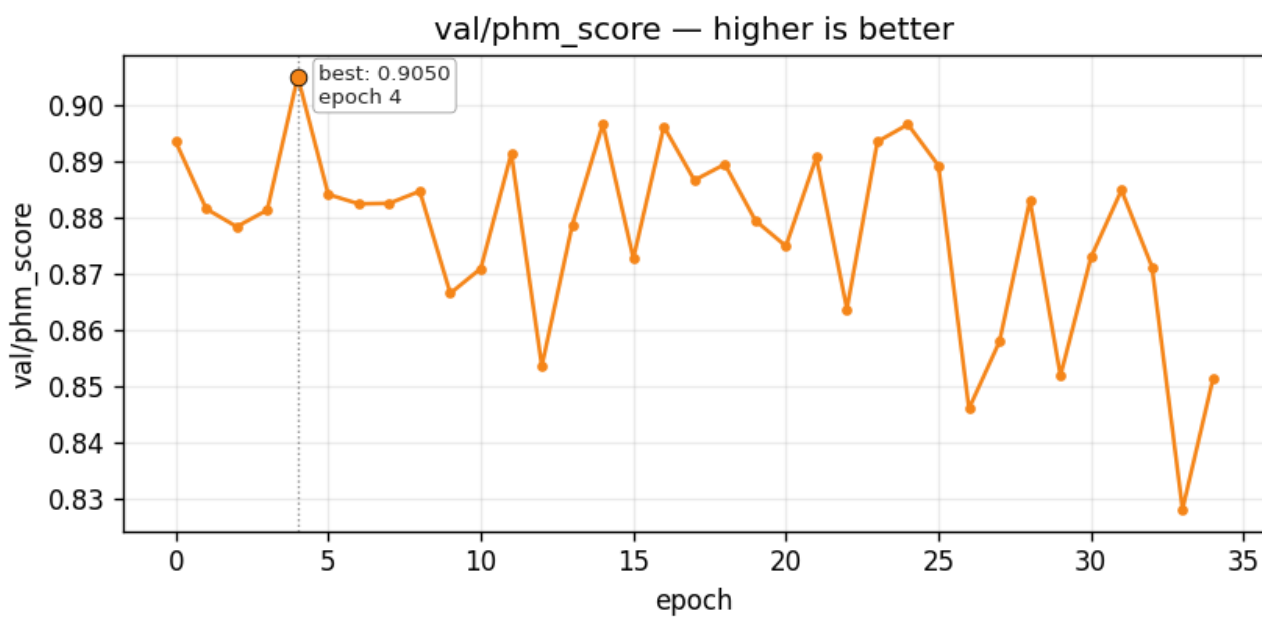


val/phm/score over epochs

Diffusion-RUL

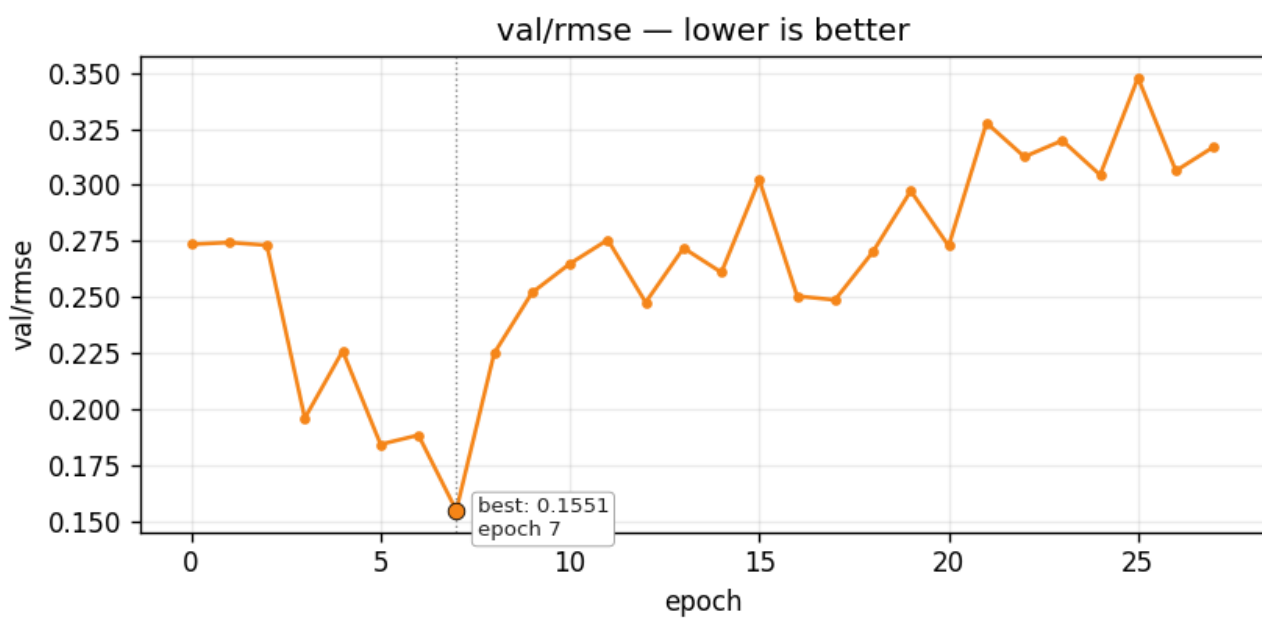


Diffusion-RUL

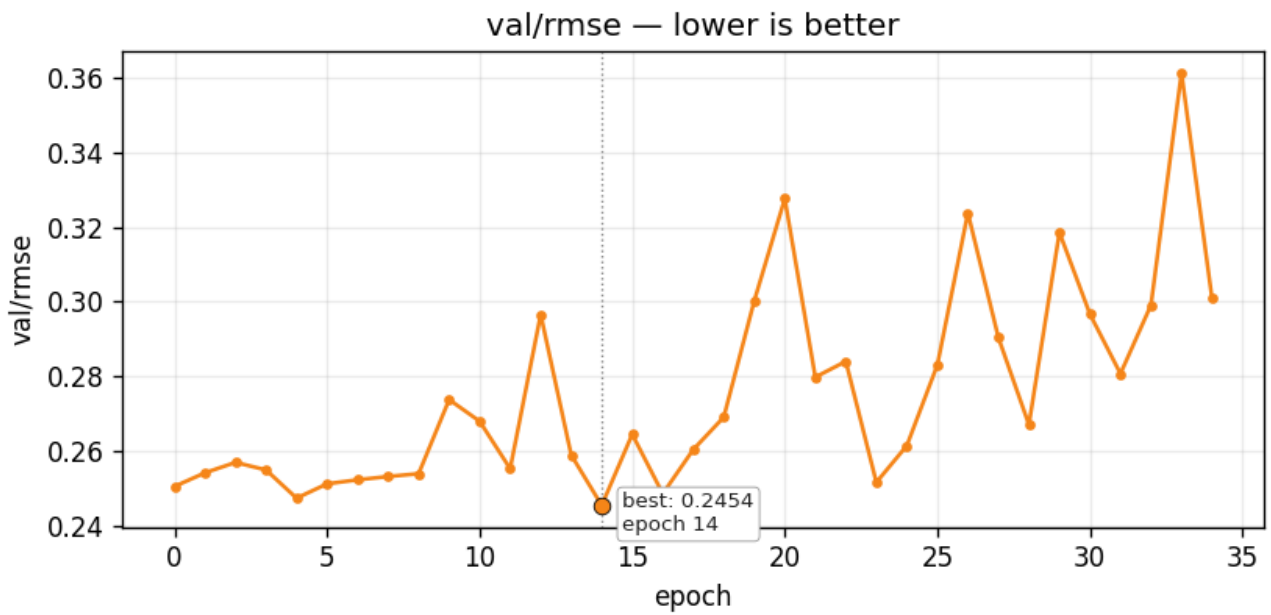


val/rmse over epochs

Diffusion-RUL



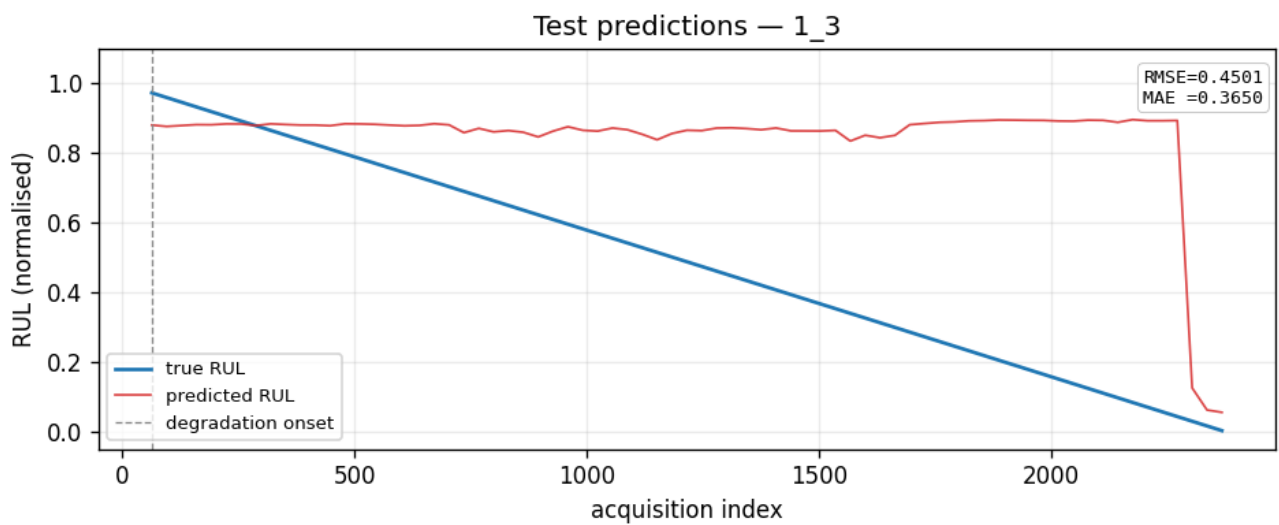
Diffusion-RUL



Per-bearing test predictions

Test bearing 1_3

Diffusion-RUL



Diffusion-RUL

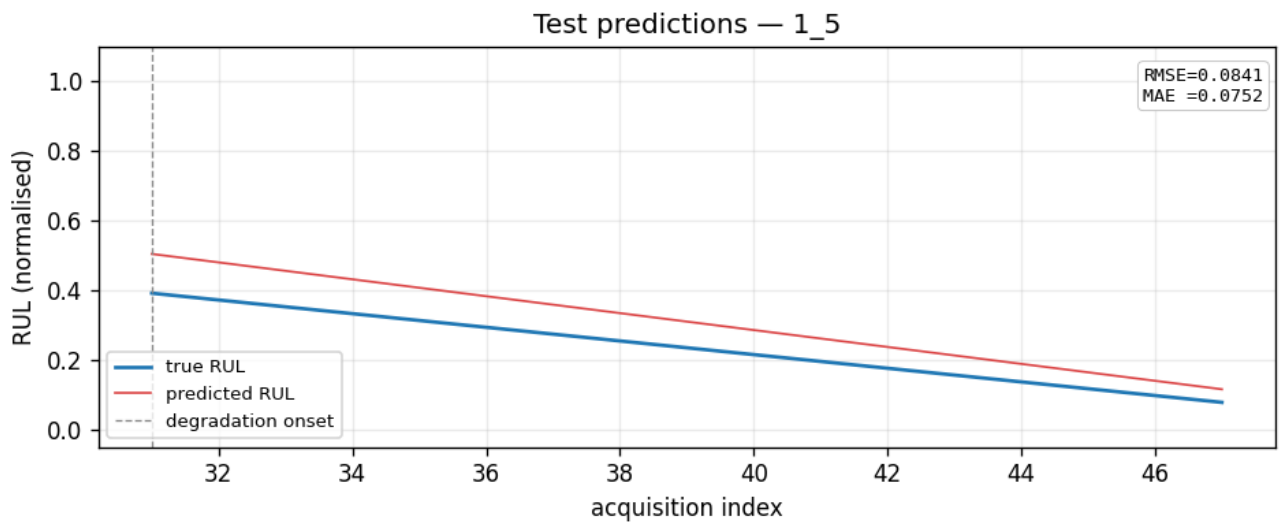
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Test bearing 1_5

Diffusion-RUL

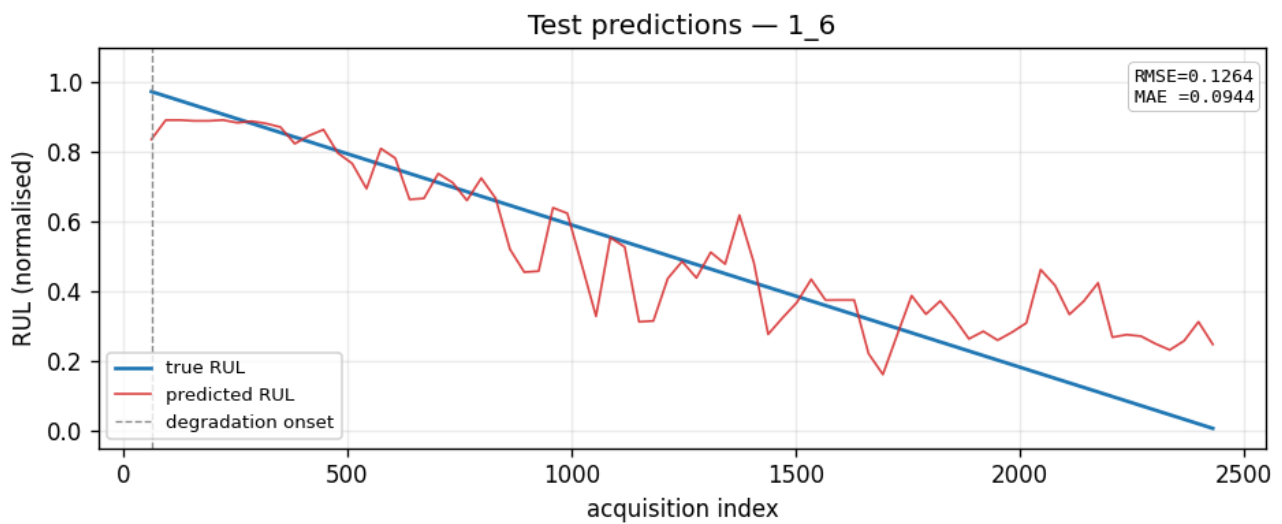
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Diffusion-RUL



Test bearing 1_6

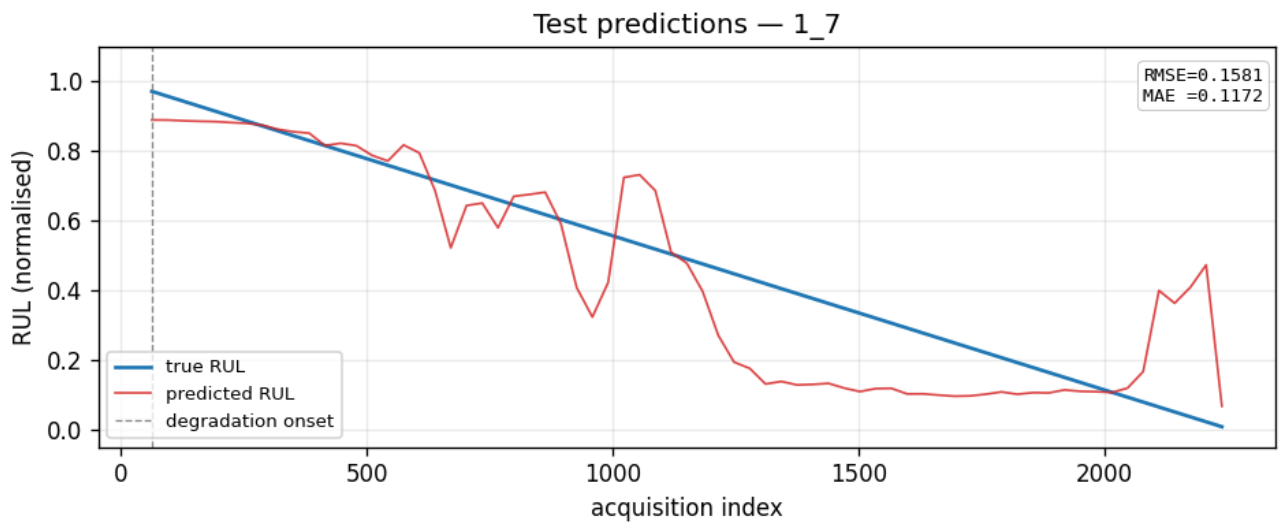
Diffusion-RUL



Diffusion-RUL

Test bearing 1_7

Diffusion-RUL

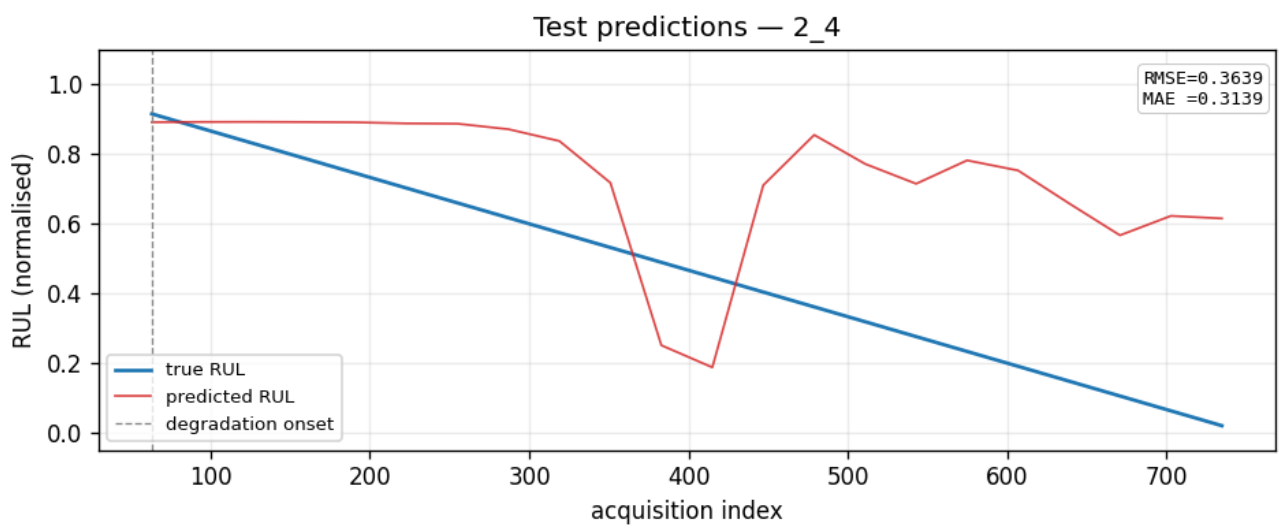


Diffusion-RUL

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Test bearing 2_4

Diffusion-RUL

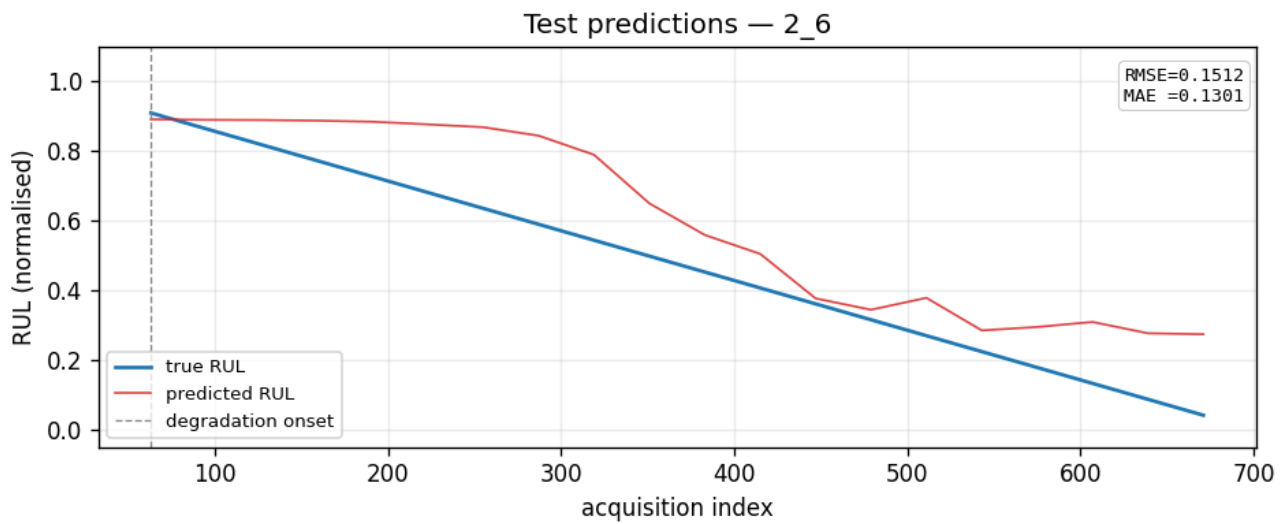


Diffusion-RUL

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Test bearing 2_6

Diffusion-RUL

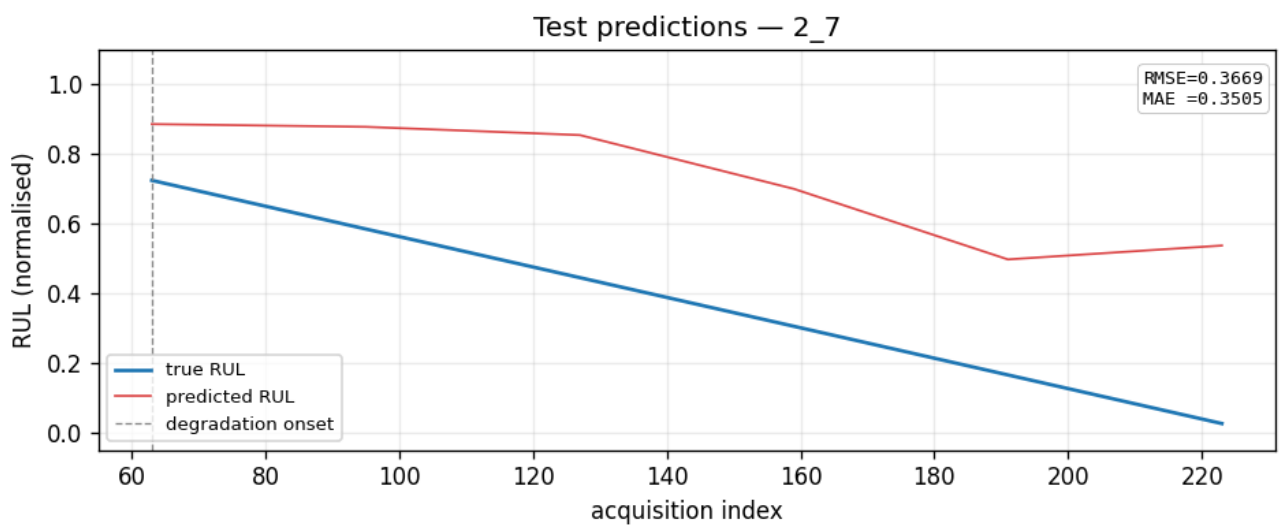


Diffusion-RUL

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Test bearing 2_7

Diffusion-RUL

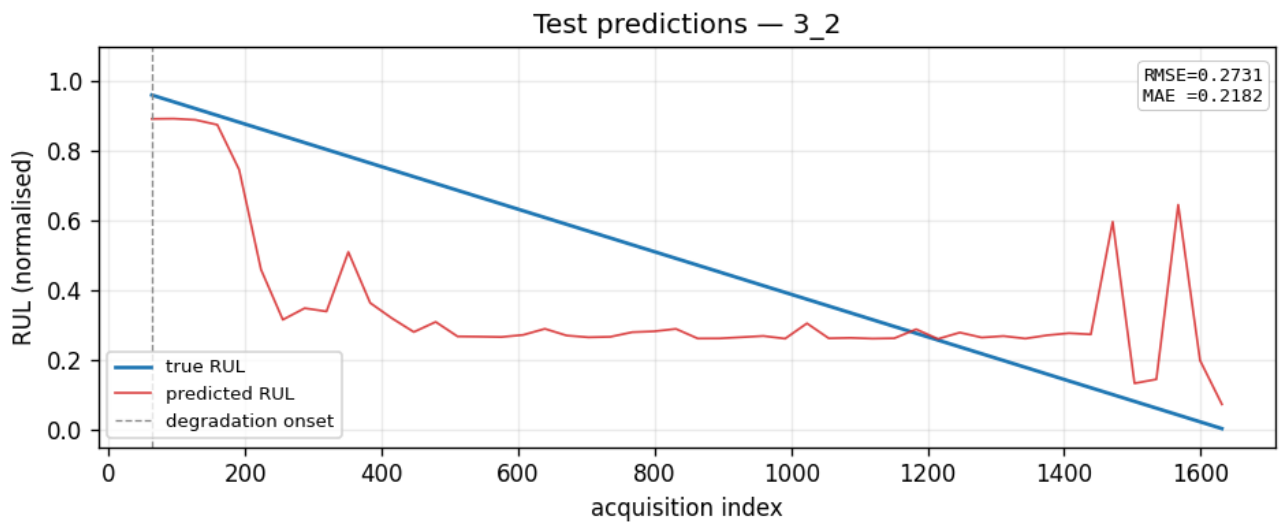


Diffusion-RUL

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Test bearing 3_2

Diffusion-RUL

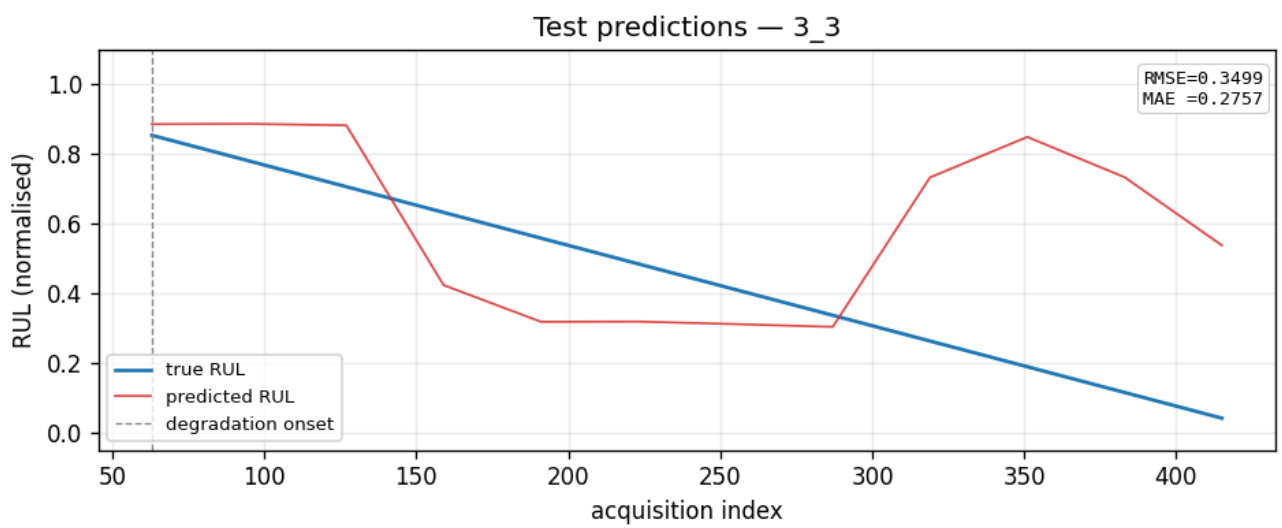


Diffusion-RUL

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Test bearing 3_3

Diffusion-RUL



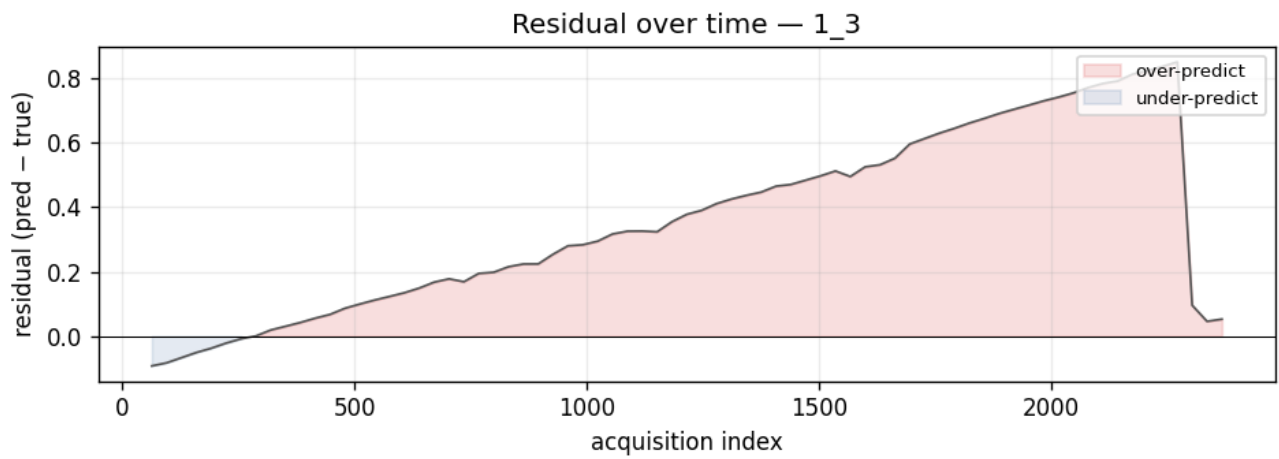
Diffusion-RUL

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Per-bearing residuals over time

Bearing 1_3

Diffusion-RUL

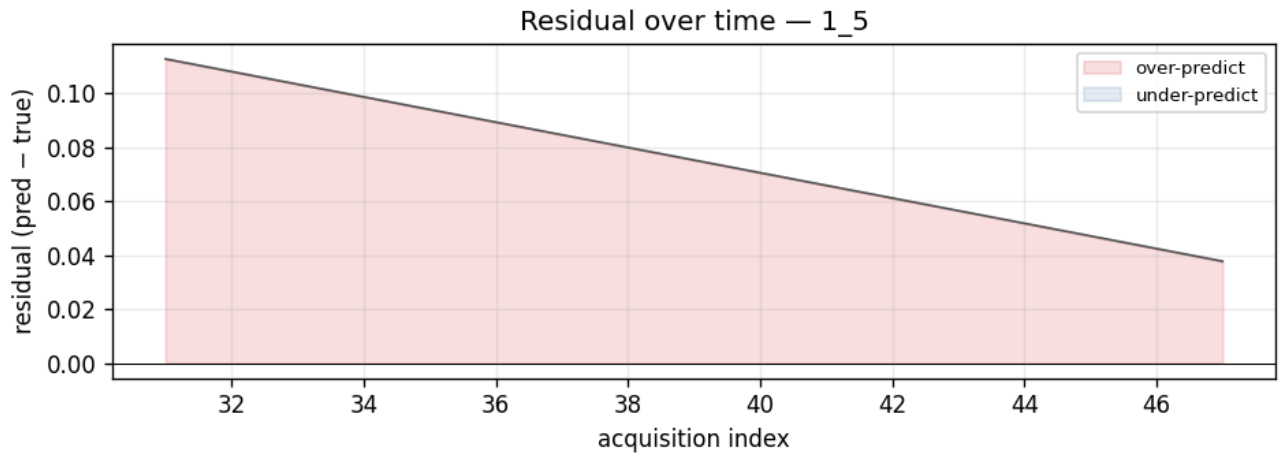


Diffusion-RUL

Bearing 1_5

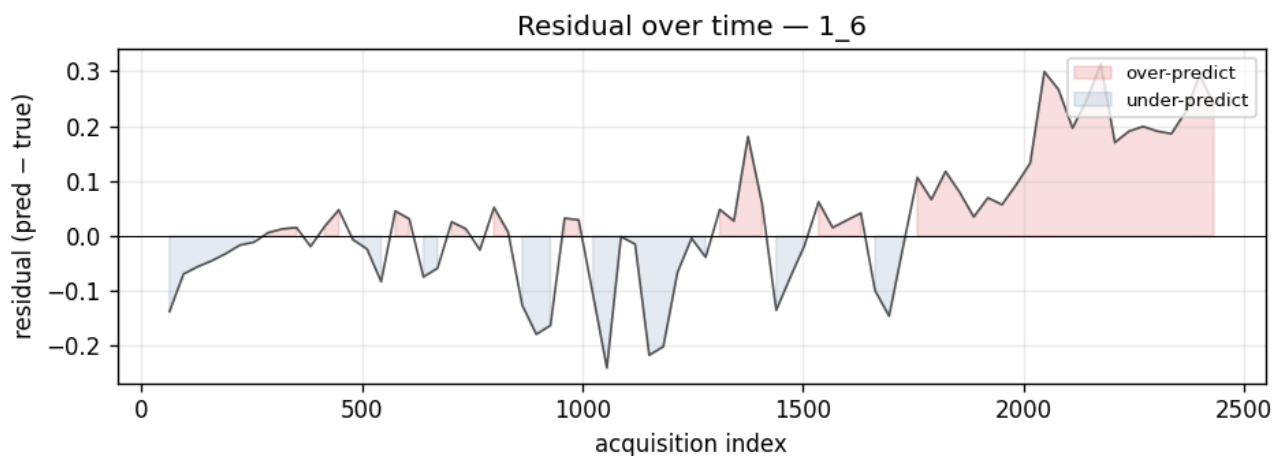
Diffusion-RUL

Diffusion-RUL



Bearing 1_6

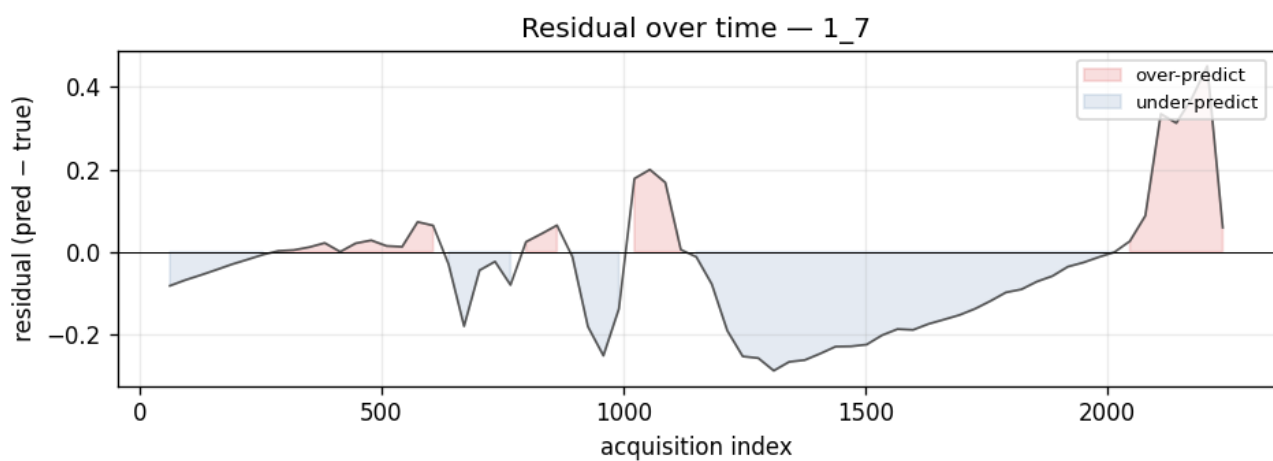
Diffusion-RUL



Diffusion-RUL

Bearing 1_7

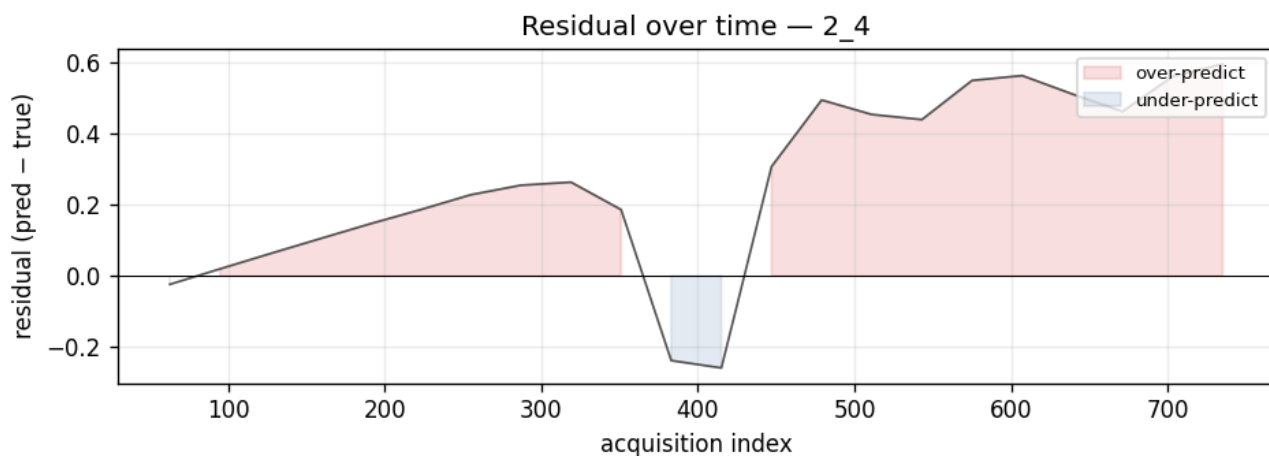
Diffusion-RUL



Diffusion-RUL

Bearing 2_4

Diffusion-RUL

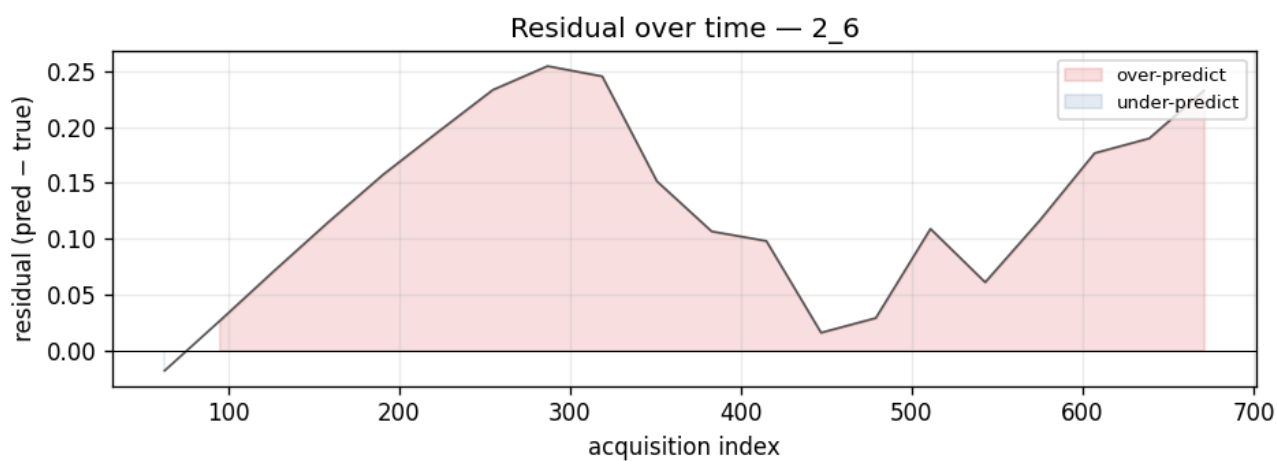


Diffusion-RUL

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Bearing 2_6

Diffusion-RUL

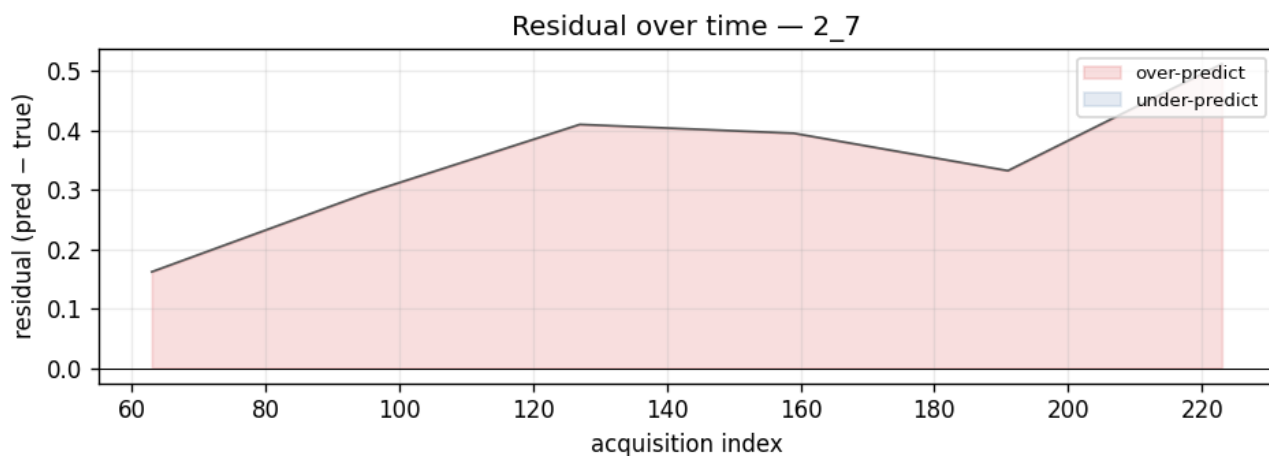


Diffusion-RUL

—

Bearing 2_7

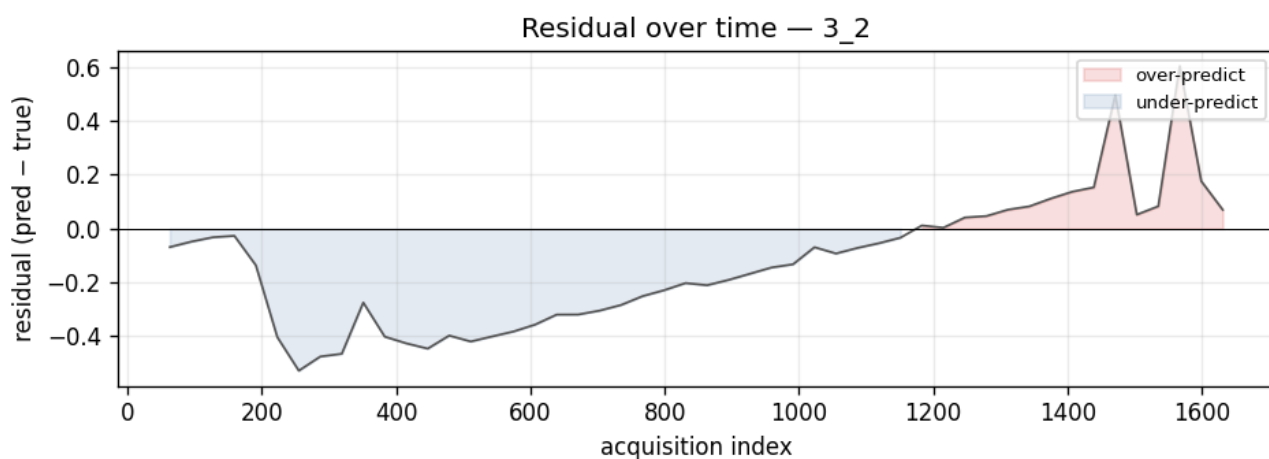
Diffusion-RUL



Diffusion-RUL

Bearing 3_2

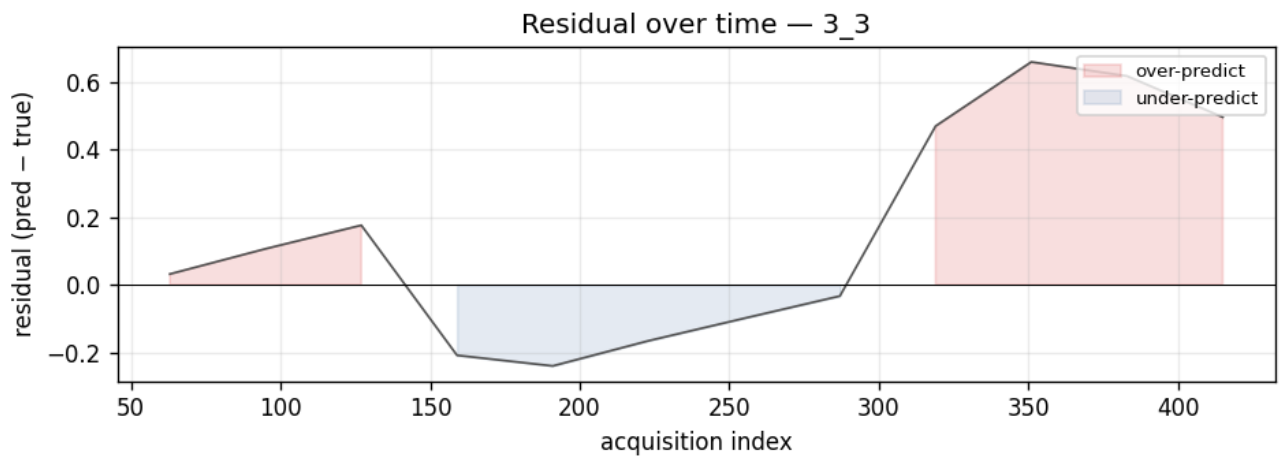
Diffusion-RUL



Diffusion-RUL

Bearing 3_3

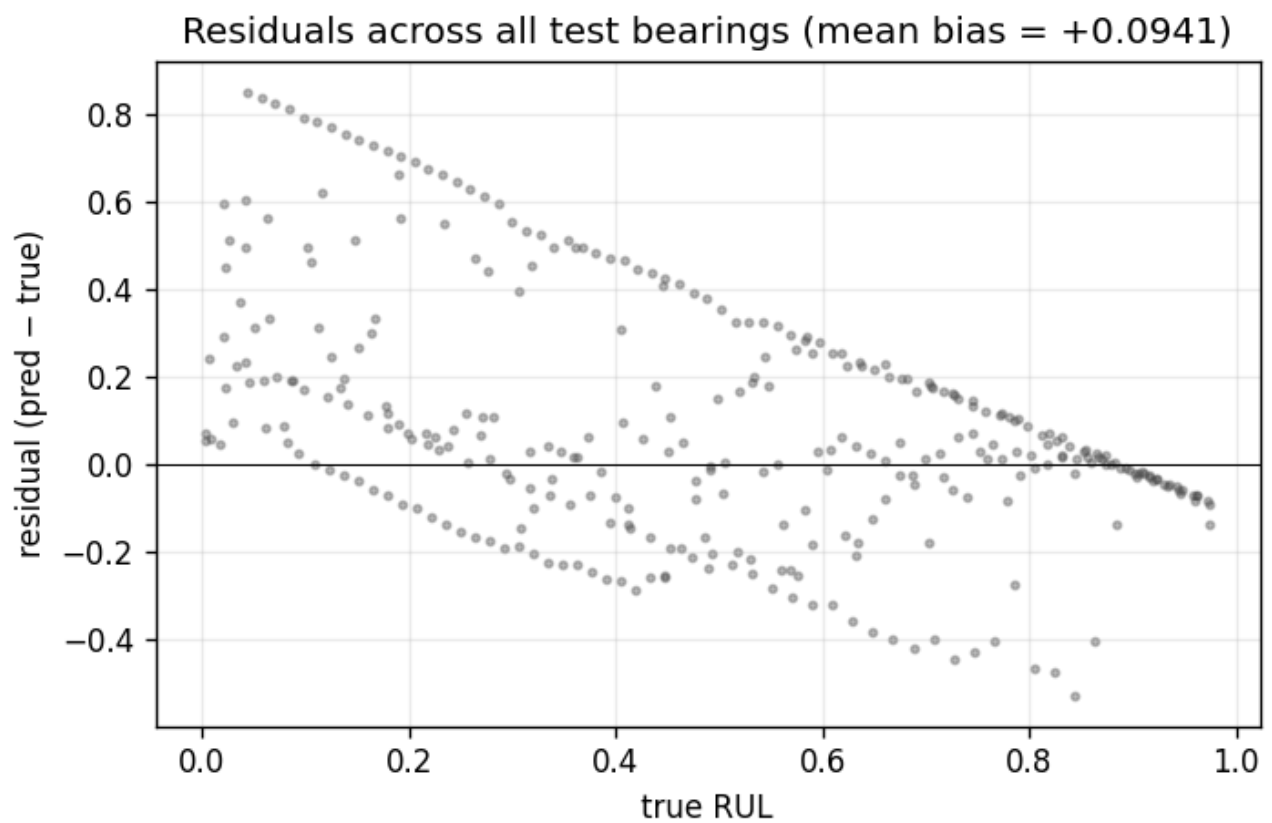
Diffusion-RUL



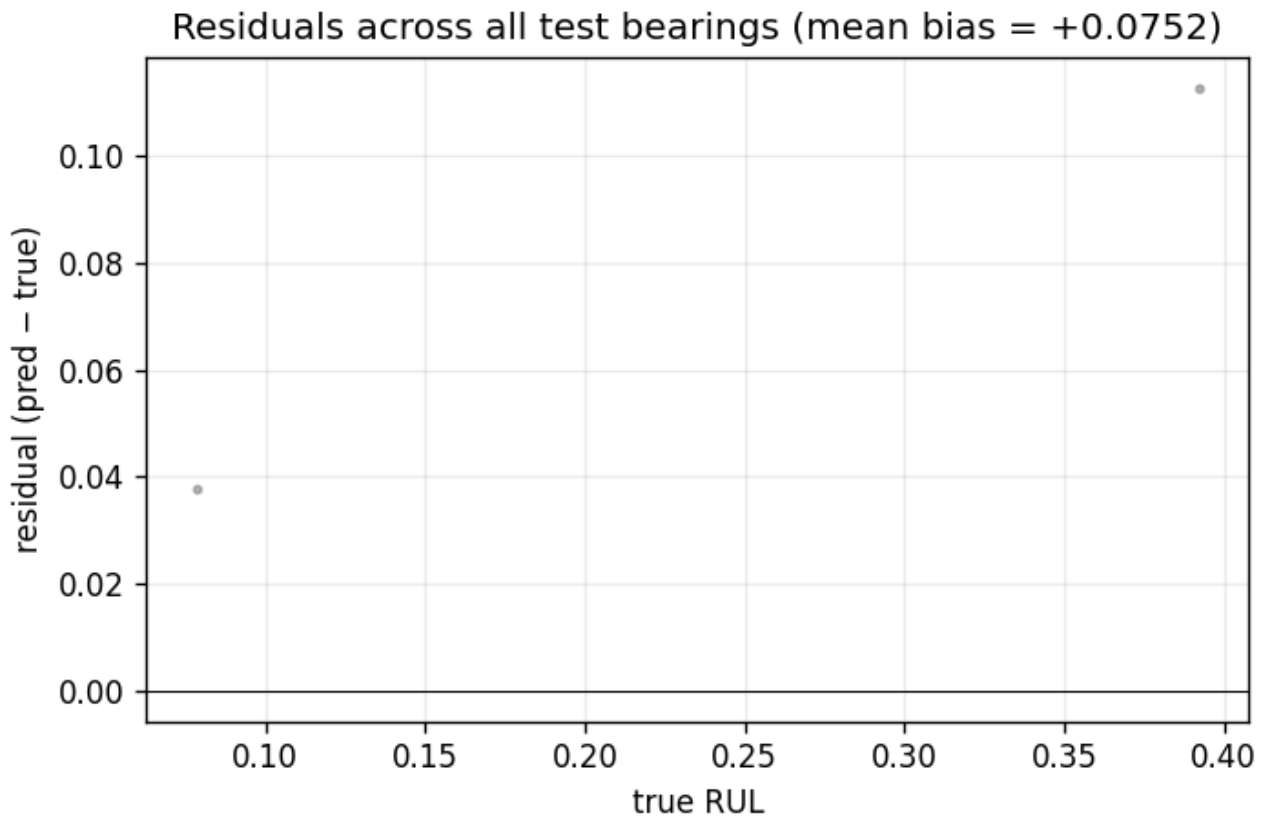
Diffusion-RUL

Residuals (all test points)

Diffusion-RUL

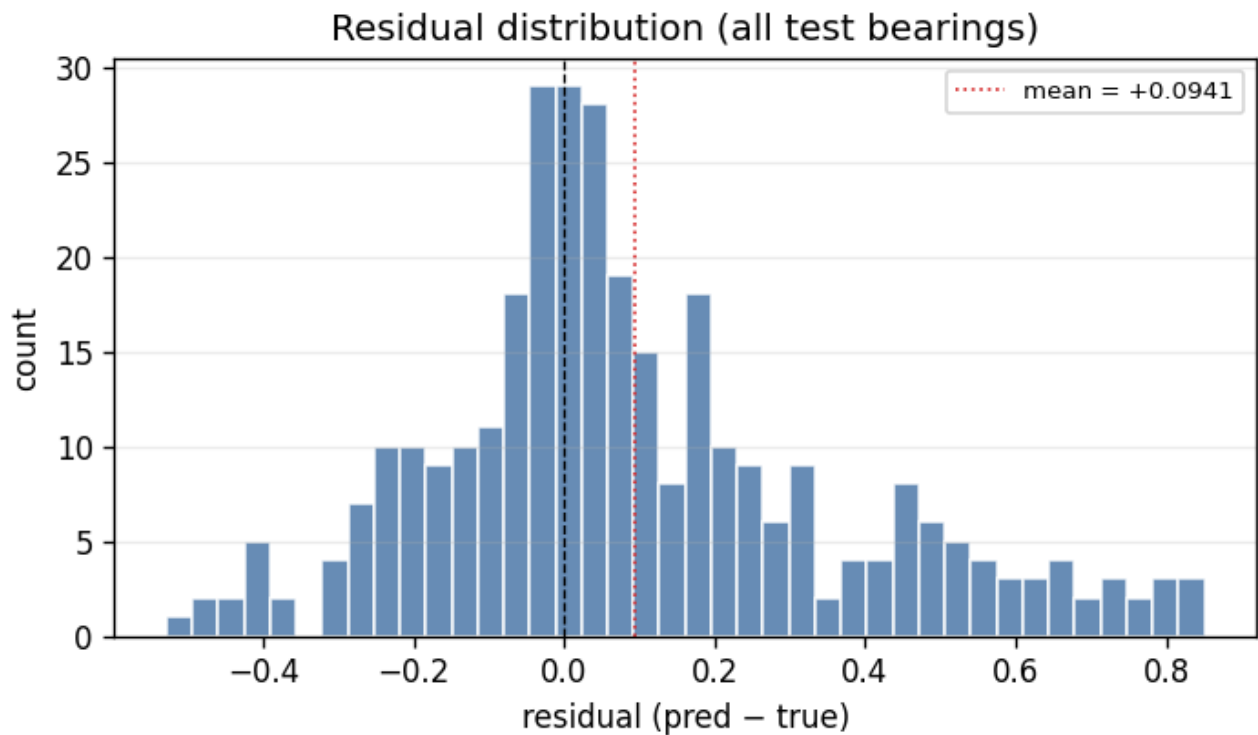


Diffusion-RUL

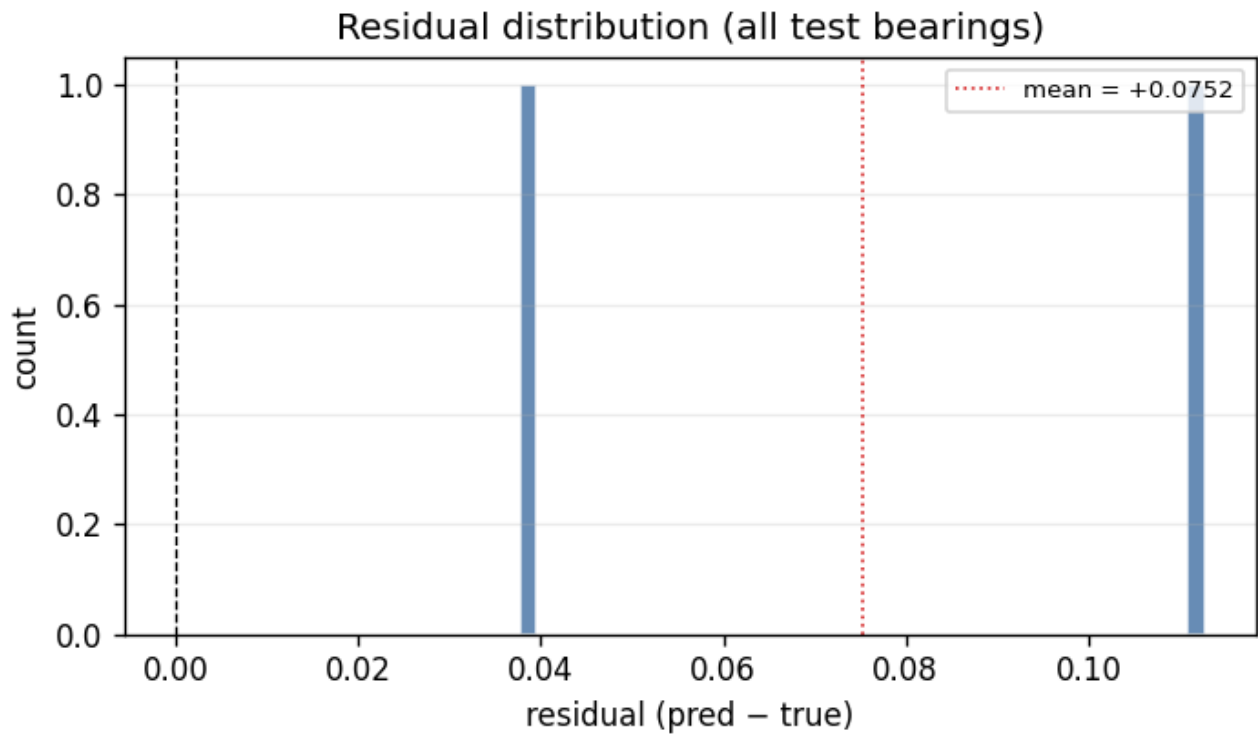


Residual histogram

Diffusion-RUL

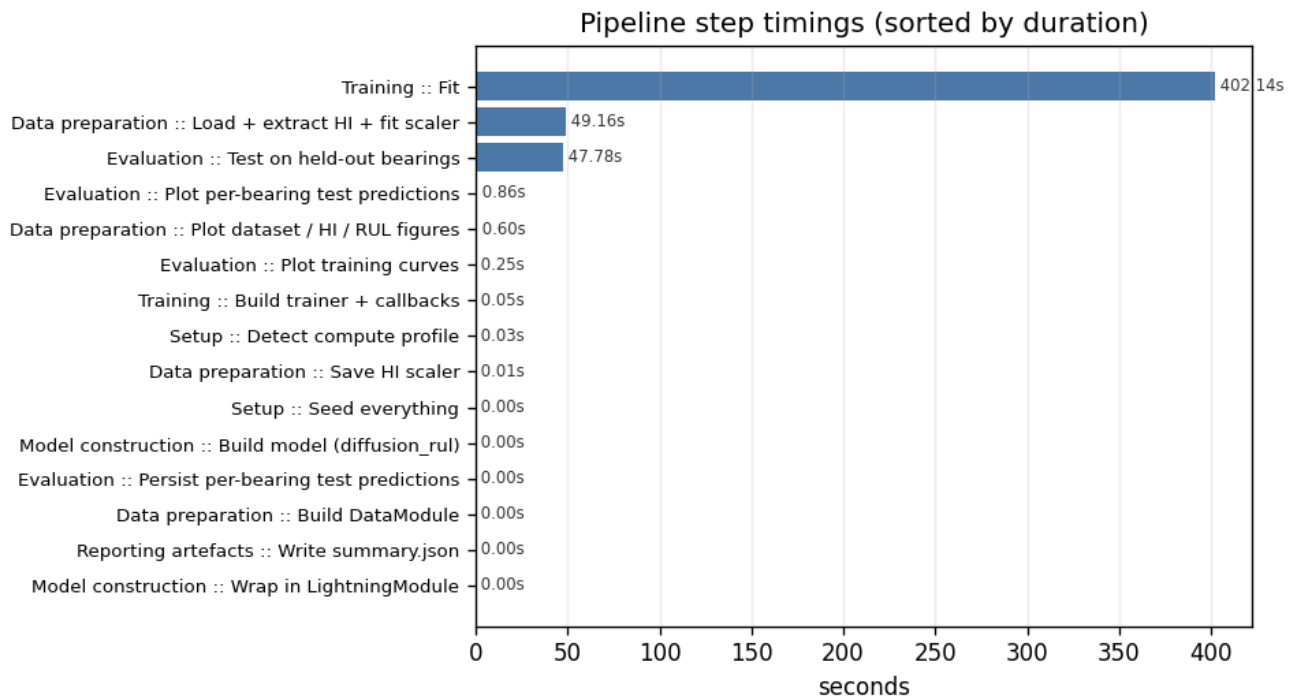


Diffusion-RUL

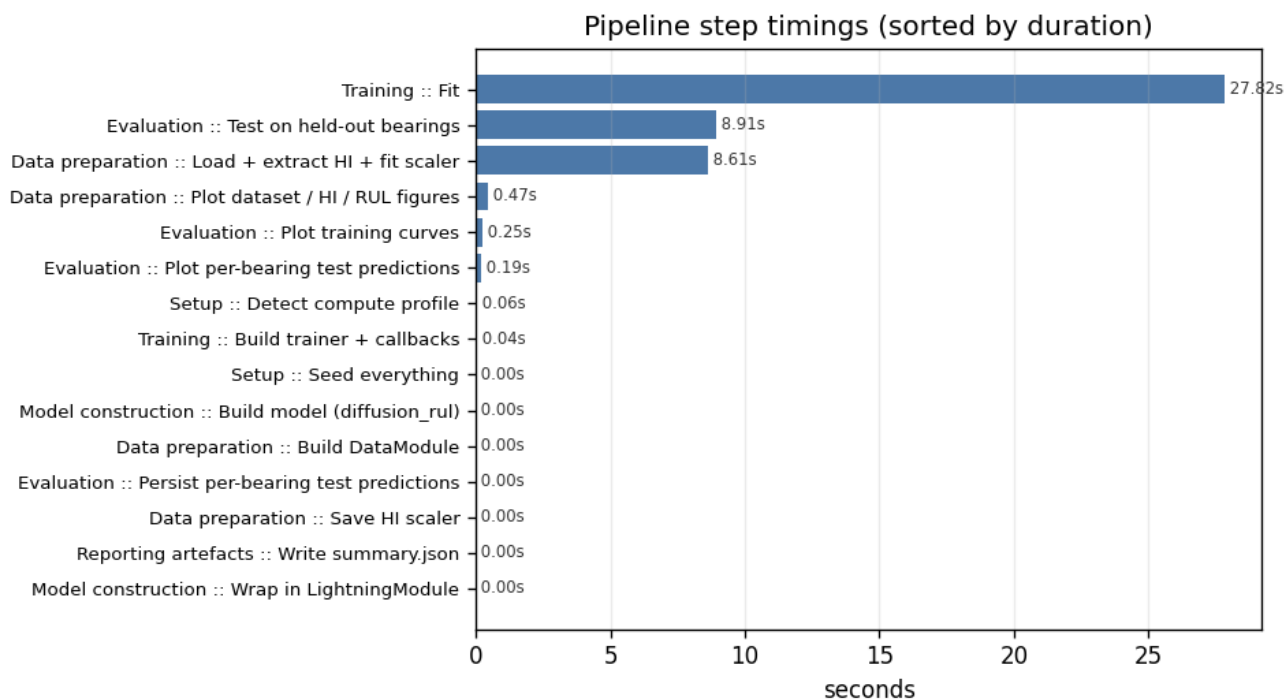


Pipeline step timings

Diffusion-RUL



Diffusion-RUL



Configuration (merged YAML)

Diffusion-RUL

```
data: dataset: phm2012 root: ../data-bearing/ieee-phm-2012 cache_dir:
../data-bearing/processed/phm2012 window_length: 64 stride_train: 1 stride_eval: 32
label_scheme: linear smoothing_alpha: 0.1 n_bands: 5 batch_size: 128 num_workers: 0
train_bearings: - '1_1' - '1_2' - '1_4' - '2_1' - '2_3' - '2_5' - '3_1' val_bearings: -
'1_5' - '2_2' test_bearings: - '1_3' - '1_6' - '1_7' - '2_4' - '2_6' - '2_7' - '3_2' -
'3_3' model: name: diffusion_rul d_model: 96 n_heads: 4 n_layers: 2 n_noise_levels: 4
max_noise_std: 0.35 dropout: 0.1 train: max_epochs: 200 lr: 0.001 weight_decay: 0.0001
scheduler: cosine warmup_epochs: 5 monotonicity_weight: 0.01 gradient_clip_val: 1.0
early_stopping_patience: 20 precision: 32-true seed: 42 results_root: results run_id: null
```

Diffusion-RUL

```
data: dataset: xjtusy root: ../data-bearing/xtju-sy cache_dir:
../data-bearing/processed/xjtusy window_length: 32 stride_train: 1 stride_eval: 16
label_scheme: linear smoothing_alpha: 0.1 n_bands: 5 batch_size: 128 num_workers: 0
train_bearings: - '1_1' - '1_2' - '1_3' - '2_1' - '2_2' val_bearings: - '1_4' - '2_3'
test_bearings: - '1_5' model: name: diffusion_rul d_model: 96 n_heads: 4 n_layers: 2
n_noise_levels: 4 max_noise_std: 0.35 dropout: 0.1 train: max_epochs: 200 lr: 0.001
weight_decay: 0.0001 scheduler: cosine warmup_epochs: 5 monotonicity_weight: 0.01
gradient_clip_val: 1.0 early_stopping_patience: 20 precision: 32-true seed: 42
results_root: results run_id: null
```